Lecture 2: the atom in modern chemistry

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/09/19

Major points of last lecture

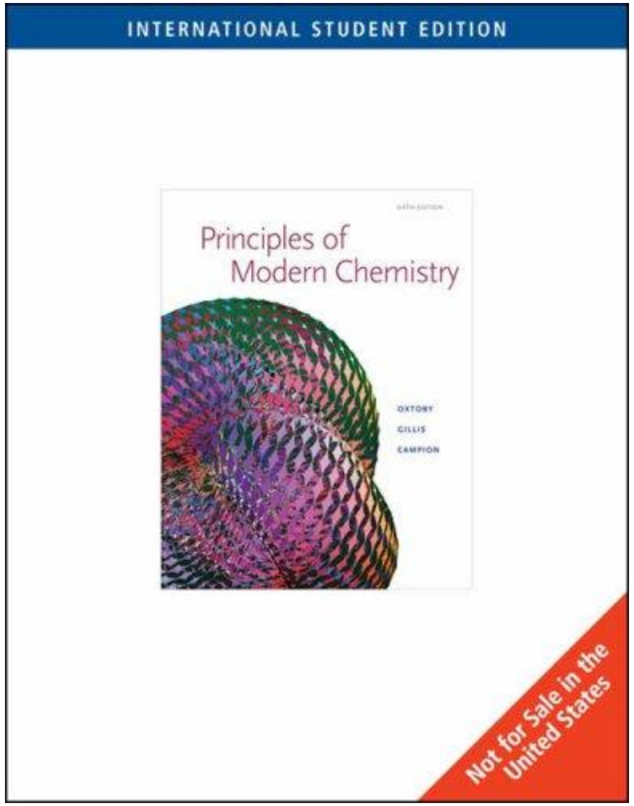
1. Discuss class syllabus

2. Establish the context of general chemistry

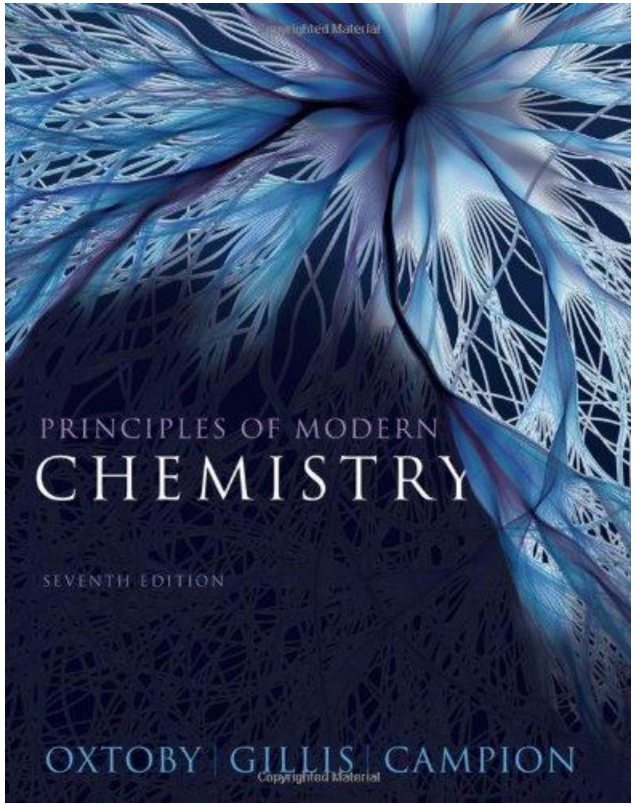
Textbooks

《Principles of Modern Chemistry》

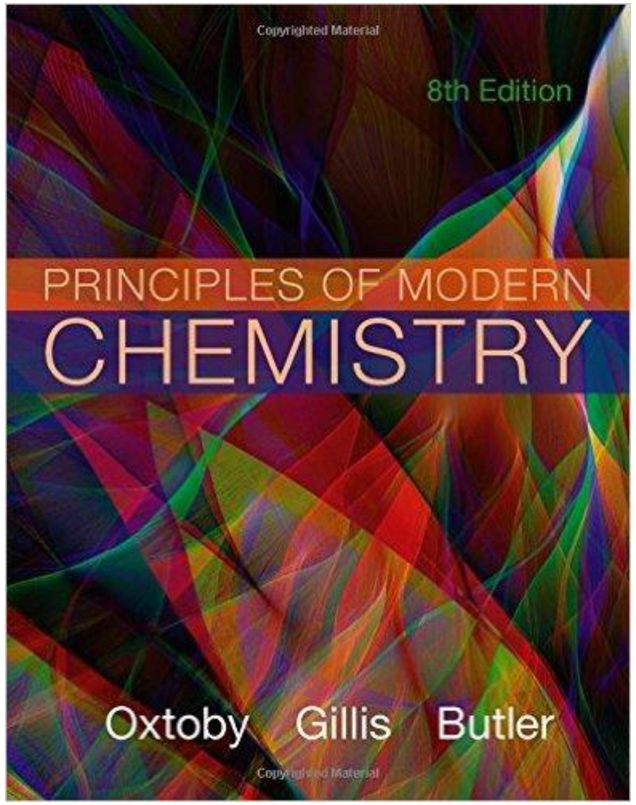
6th Int'l Edition, 2007
956 pages



7th Edition, 2012
1132 pages



8th Edition, 2015
1136 pages



Week	Monday	Wednesday	Friday
1	subject to change	Sept 17 Overview & Introduction	Sept 19 The atom in modern chemistry
2	Sept 22 Recitation		Sept 26 Chemical formulas, equations, and reaction yields Quiz 1
3	Sept 29 PS1 due, Recitation	Oct 1 holiday	Oct 3 holiday
4	Oct 6 holiday		Oct 10 Atomic shells and classical models of chemical Quiz 2
5	Oct 13 Recitation	Oct 15 Atomic shells and classical models of chemical	Oct 17 Introduction to quantum mechanics-1 Quiz 3
6	Oct 20 PS2 due, Recitation		Oct 24 Introduction to quantum mechanics-2 Review 1 Quiz 4
7	Oct 27 Recitation	Oct 29 Midterm 1	Oct 31 Quantum mechanics and atomic structure-1 Quiz 5
8	Nov 3 PS3 due, Recitation		Nov 7 Quantum mechanics and atomic structure-2 Quiz 6
9	Nov 10 Recitation	Nov 12 Quantum mechanics and molecular structure-1	Nov 14 Quantum mechanics and molecular structure-2 Quiz 7
10	Nov 17 PS4 due, Recitation		Nov 21 Quantum mechanics and molecular structure-3 Quiz 8
11	Nov 24 Recitation	Nov 26 Quantum mechanics and molecular structure-4	Nov 28 Quantum mechanics and molecular structure-5 Quiz 9
12	Dec 1 PS5 due, Recitation		Dec 5 Bonding in organic molecules Review 2 Quiz 10
13	Dec 8 Recitation	Dec 10 Midterm 2	Dec 12 Bonding in transition metal compounds and coordination complexes-1 Quiz 11
14	Dec 15 PS6 due, Recitation		Dec 19 Bonding in transition metal compounds and coordination complexes-2 Quiz 12
15	Dec 22 Recitation	Dec 24 Gas, solids, liquid, and phase transitions	Dec 26 Rare earths Quiz 13
16	Dec 29 PS7 due, Recitation		Jan 2 Spectroscopy Review 3 Quiz 14

Class syllabus

Quiz x13: 26%

- 随堂测验一共14次，计分时舍弃一个最低分，每次15-25分钟

Problem set x7: 28%

- 周五课后发布
- 下一个周一习题课前（下午三点）提交至bb系统，每次总分为 4 分；周二上午10点前补交，每次总分为 3 分（即实际成绩乘以 3/4）；学期结束前补交，每次总分为 2 分
- 学生之间可以就课后习题进行讨论，但不允许相互抄袭，建议先充分讨论然后再开始单独动笔解答。任课老师和助教有权将疑似抄袭的作业提交学校的有关委员会进行鉴定

Project: 6%

Mid-term exam x2: 20%

- 因故无法参加期中考试的学生需按照学校规定请假

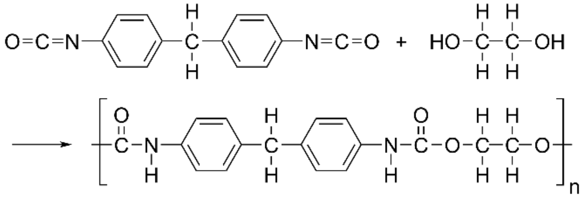
Final exam x1: 20%

- 17-18周进行

Outline

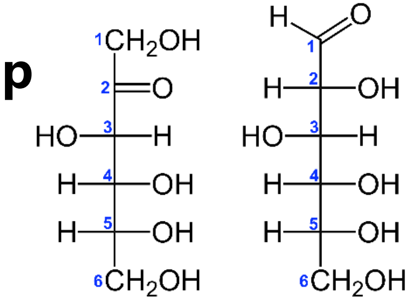
- Why is chemistry central to our life?
- What does chemistry study?
- What has chemistry achieved and what hasn’t?
- What do we benefit from learning chemistry?

衣



食

High-fructose corn syrup



住

Calcite: $CaCO_3$
Dolomite: $CaMg(CO_3)_2$

行

Cathode: NCA ($LiNi_xCo_yAl_zO_2$)
Anode: graphite (C)

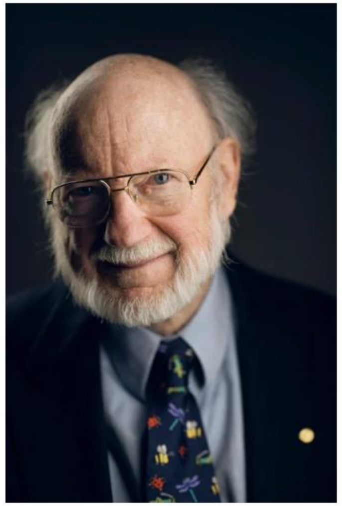
Outline

- Why is chemistry central to our life?
- What does chemistry study?
- What has chemistry achieved and what hasn’t?
- What do we benefit from learning chemistry?

What does chemistry study?

- Create entirely new substances designed to have specific properties
- Properties of substances
- The reactions that transform substances into other substances
- Understanding how and why specific chemical reactions occur
- Tailor the properties of existing substances to meet particular needs

The Nobel Prize in Physiology or Medicine 2015



William C. Campbell

Prize share: 1/4

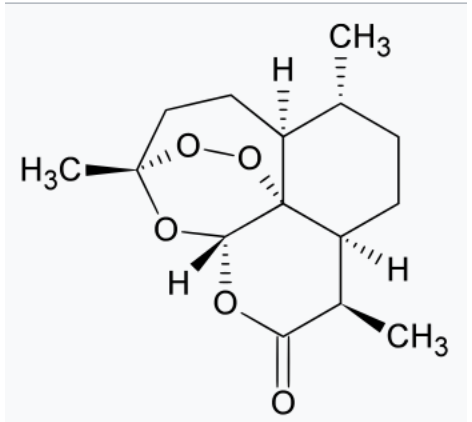
Satoshi Ōmura

Prize share: 1/4

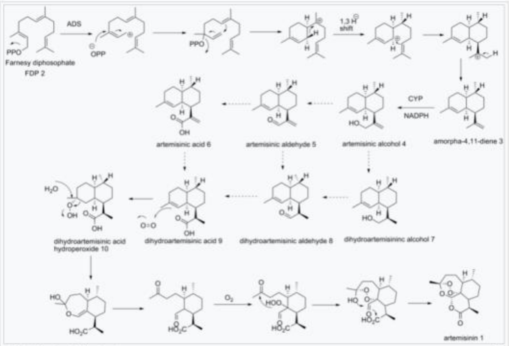
Tu Youyou

Prize share: 1/2

青蒿素

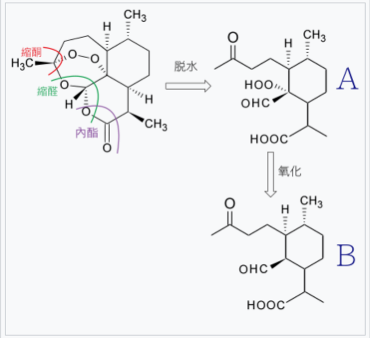


青蒿素在黄花蒿中的生物合成



青蒿素的生物合成

全合成



青蒿素全合成的逆合成分析

微生物工程菌全合成青蒿酸

The Nobel Prize in Physiology or Medicine 2015 **was divided, one half jointly to William C. Campbell and Satoshi Ōmura "for their discoveries concerning a novel therapy against infections caused by roundworm parasites" and the other half to <u>Tu Youyou "for her discoveries concerning a novel therapy against Malaria"</u>**

https://www.nobelprize.org/prizes/medicine/2015/summary/

Major points of today's lecture

- The atom in modern chemistry
  1. The nature of modern chemistry
  2. Elements: the building blocks of matter
  3. Laws of chemical combination
  4. The physical structure of atoms
  5. Mass spectrometer and isotopes
  6. The mole

History of chemistry

构成物质的基本单元?

西方“四元素”

- Earth, Water, Air, and Fire

东方“五行”

- 金、木、水、火、土

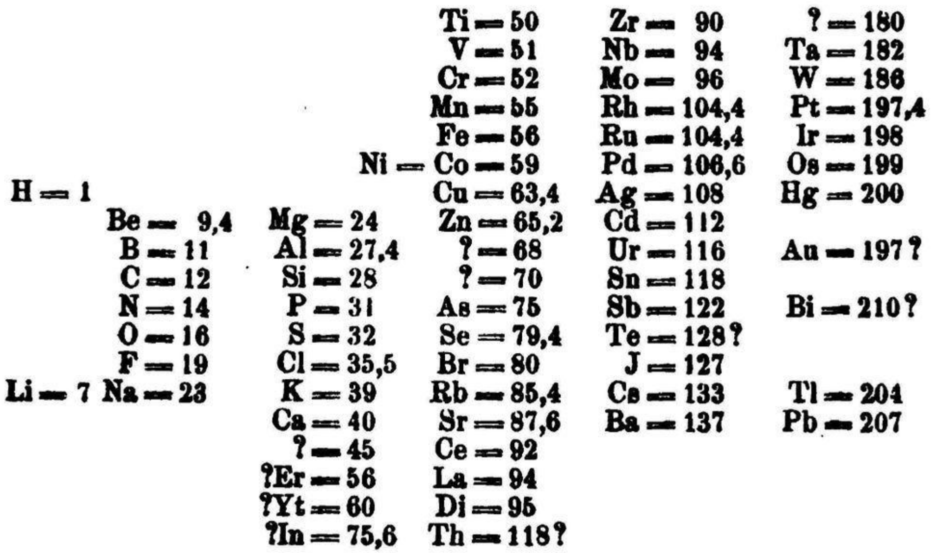
History of chemistry

- **Alchemy (炼金术) and alchemists.**

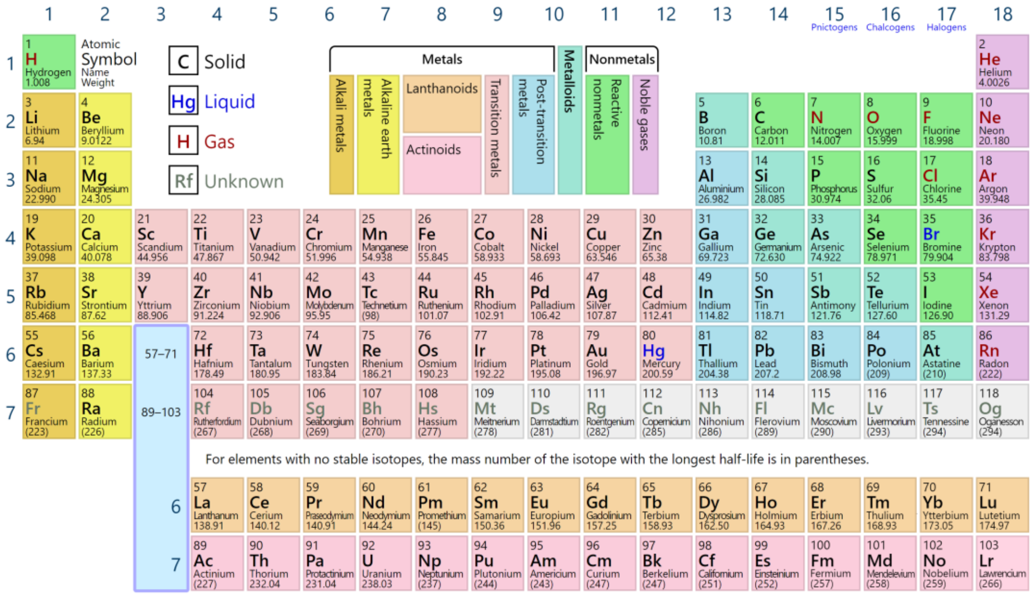
yellow color + softness + ductility = gold?

Elixir，长生不老药

History of chemistry

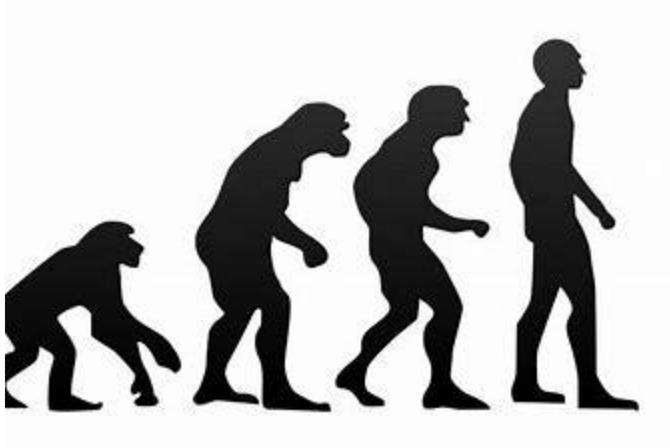


1869, Dimitri Mendeleev



2019

Human: 25 chemical elements



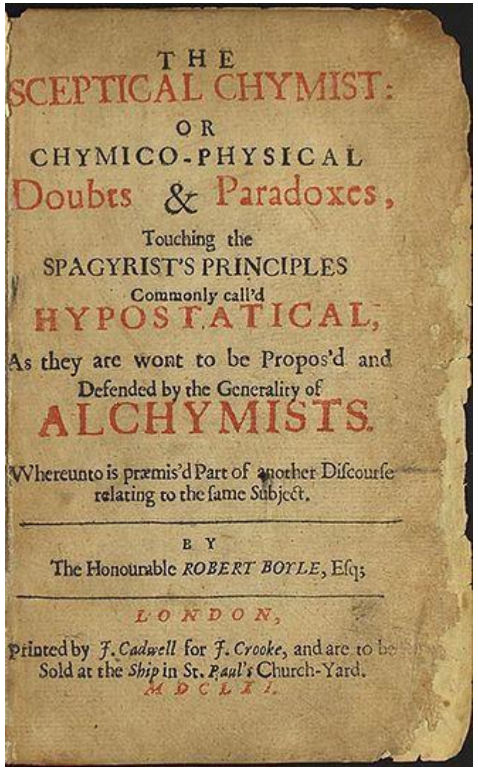
Earth: 118 chemical elements

History of chemistry

Aristotelian theory of the four elements (earth, air, fire, and water)

Boyle rejected the Aristotelian theory.

Robert Boyle(罗伯特·波义耳), 1627-1691



THE
SCEPTICAL CHYMIST:
OR
CHYMICO-PHYSICAL
Doubts & Paradoxes,
Touching the
SPAGYRIST'S PRINCIPLES
Commonly call'd
HYPOSTATICAL;
As they are wont to be Propos'd and
Defended by the Generality of
ALCHYMISTS.
Whereunto is præmis'd Part of another Discourse
relating to the same Subject.

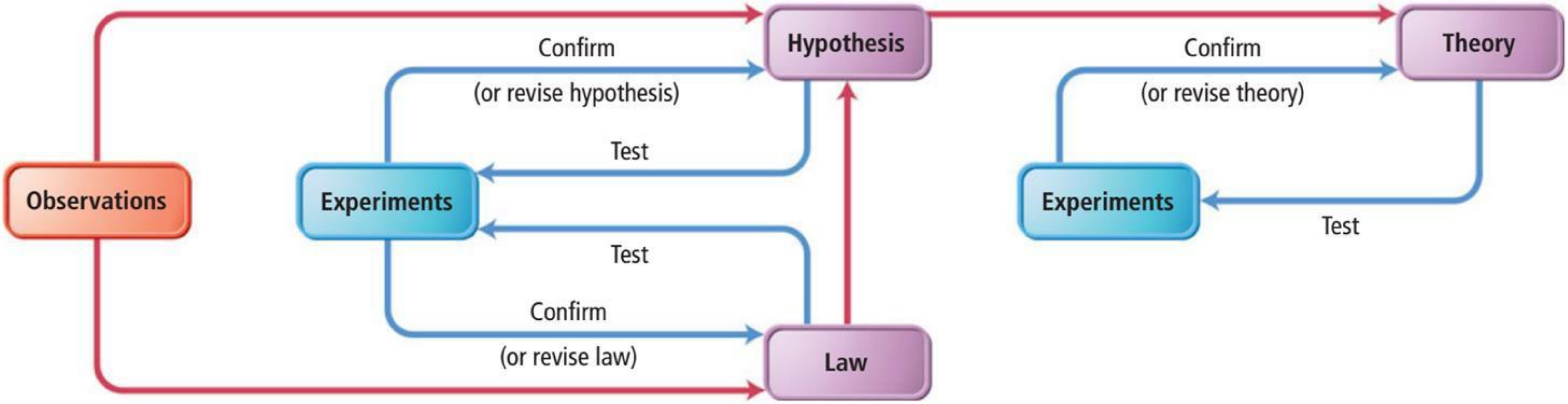
BY
The Honourable *ROBERT BOYLE*, Esq;

LONDON,
Printed by *J. Cadwell* for *J. Crooke*, and are to be
Sold at the *Ship* in St. *Paul's* Church-Yard.
MDCLXI.

The Skeptical Chemist（怀疑派化学家）, 1660s

Scientific Method

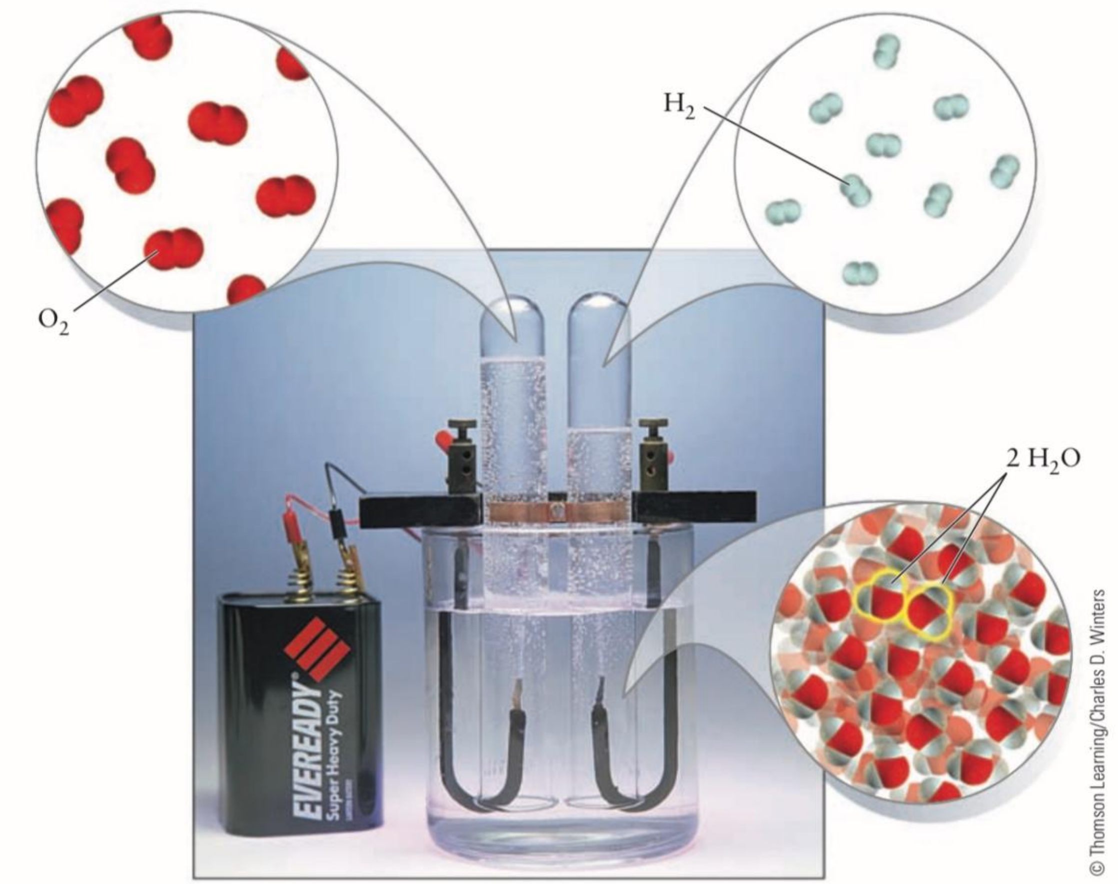
Hypothesis (假设) vs. scientific law (科学定律)



Macroscopic and Nanoscopic Methods

Macroscopic: 1mm - 1m

Nanoscopic: 1 nm



The chemical transformation, induced by the macroscopic apparatus, proceeds by rearrangement of atoms at the nanoscale.

Major points of today's lecture

- The atom in modern chemistry
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  3. Laws of chemical combination
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Elements: the building blocks of matter

element *noun* (PART)

B2 [C]

a part of something

部分；部件；要素

- *List the elements that make up a perfect dinner party.*
  列举出尽善尽美的晚宴所有的构成要素。
- *The movie had all the elements of a good thriller.*
  这部电影具备了一部精彩的惊险片的一切要素。

element *noun* (SIMPLE SUBSTANCE)

B2 [C]

a simple substance that cannot be reduced to smaller chemical parts

元素

- *Aluminium is an element.*
  铝是一种元素。

element *noun* (EARTH, AIR, ETC.)

[C]

earth, air, fire, and water from which people in the past believed everything else was made

四大元素（旧时人们认为构成一切事物的土、风、火、水）

- **Elements (单质) are substances that cannot be decomposed into two or more simpler substances by ordinary physical or chemical means. The word ordinary excludes the processes of radioactive decay, whether natural or artificial, and high-energy nuclear reactions that do transform elements into one another.**
- **Chemical elements (化学元素)**

Elements: the building blocks of matter

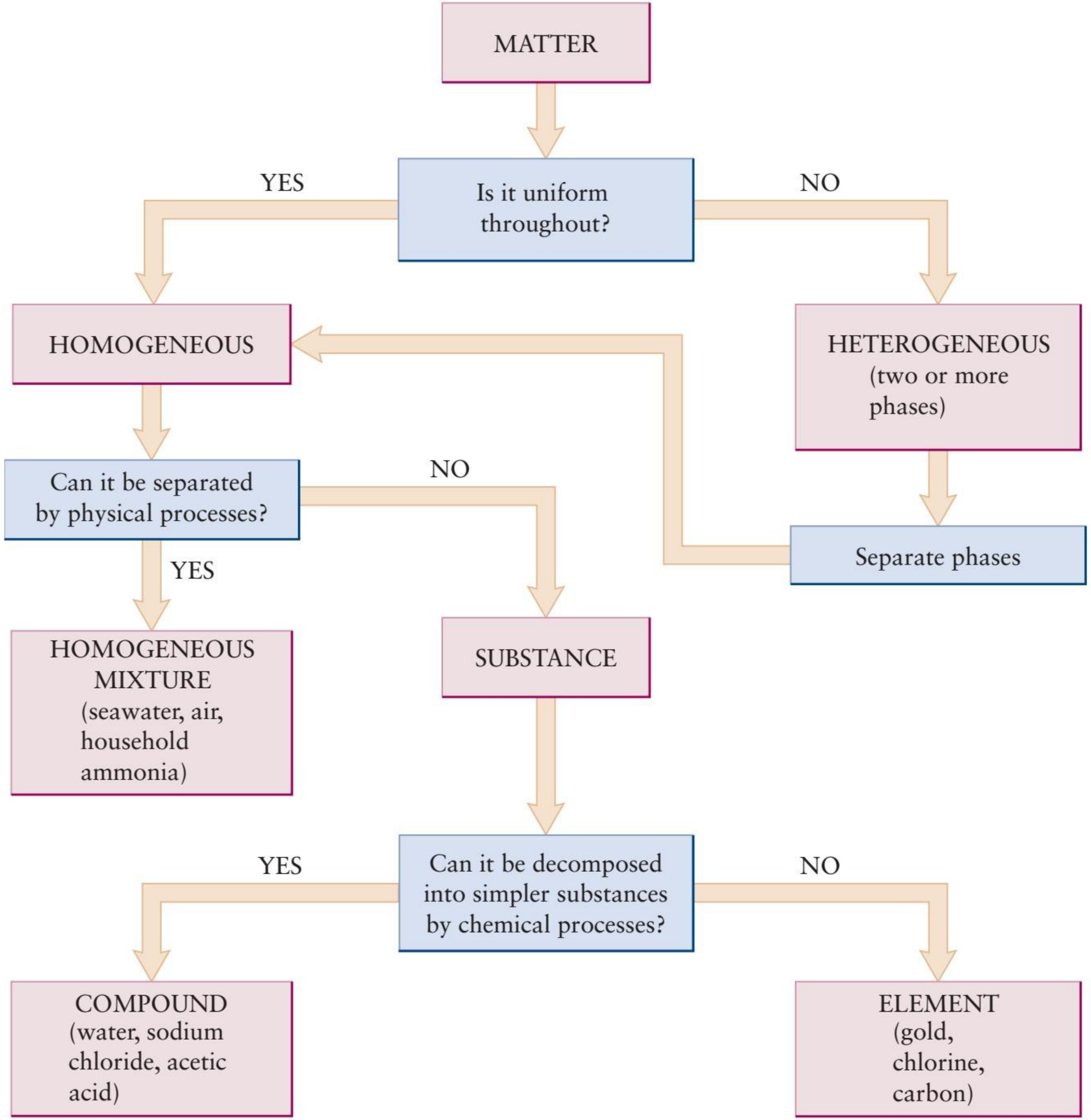
- **Chemistry: how <u>sets of pure substances</u> transform into <u>other sets of pure substances</u> in chemical reactions.**

↓

- **<u>Analysis</u> (taking things apart); <u>Synthesis</u> (putting thins together).**

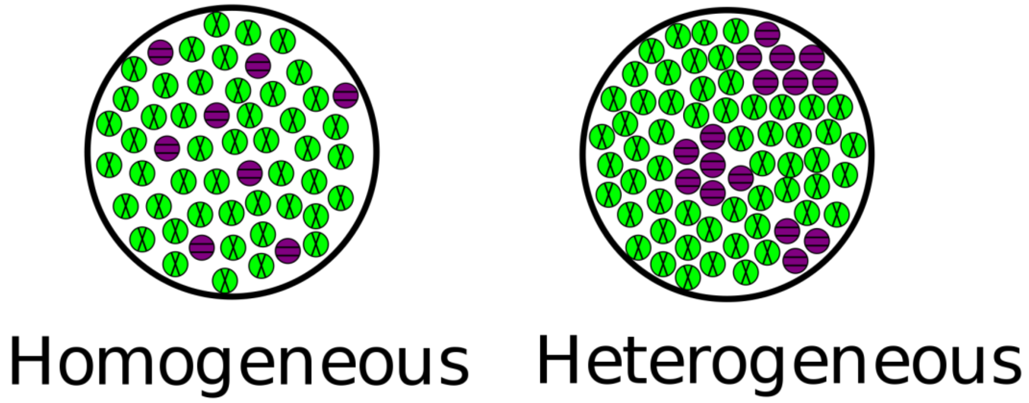
↓

Q1：what are the building blocks of matter (elements) in modern chemistry?

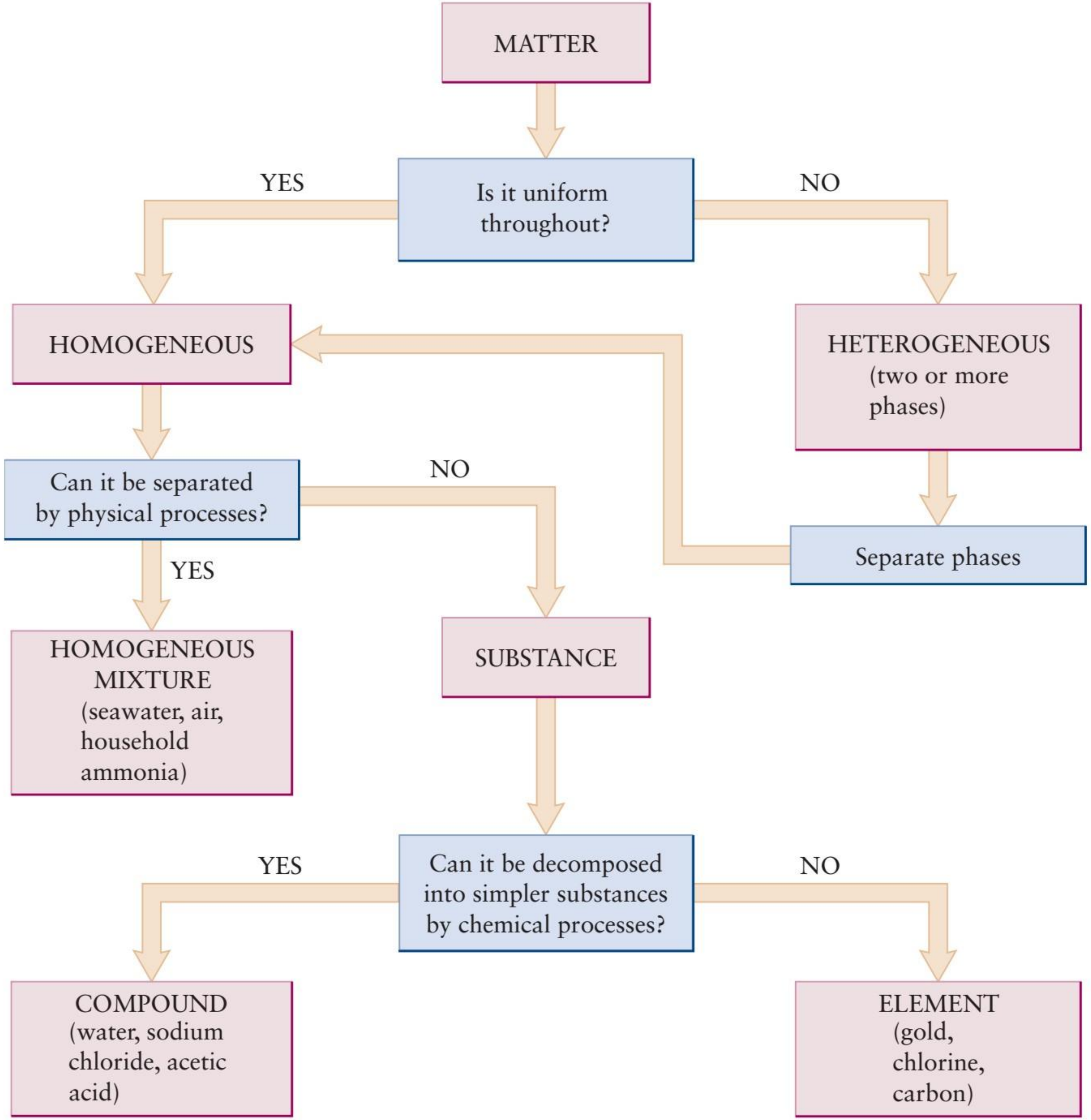


A **homogeneous (同质) material** is a solid, liquid, or gaseous that has the same proportions of its components throughout any given sample.

A **heterogeneous (异质) material** has components whose proportions vary throughout the sample.



Q1：木头? **Q2：盐开水?**

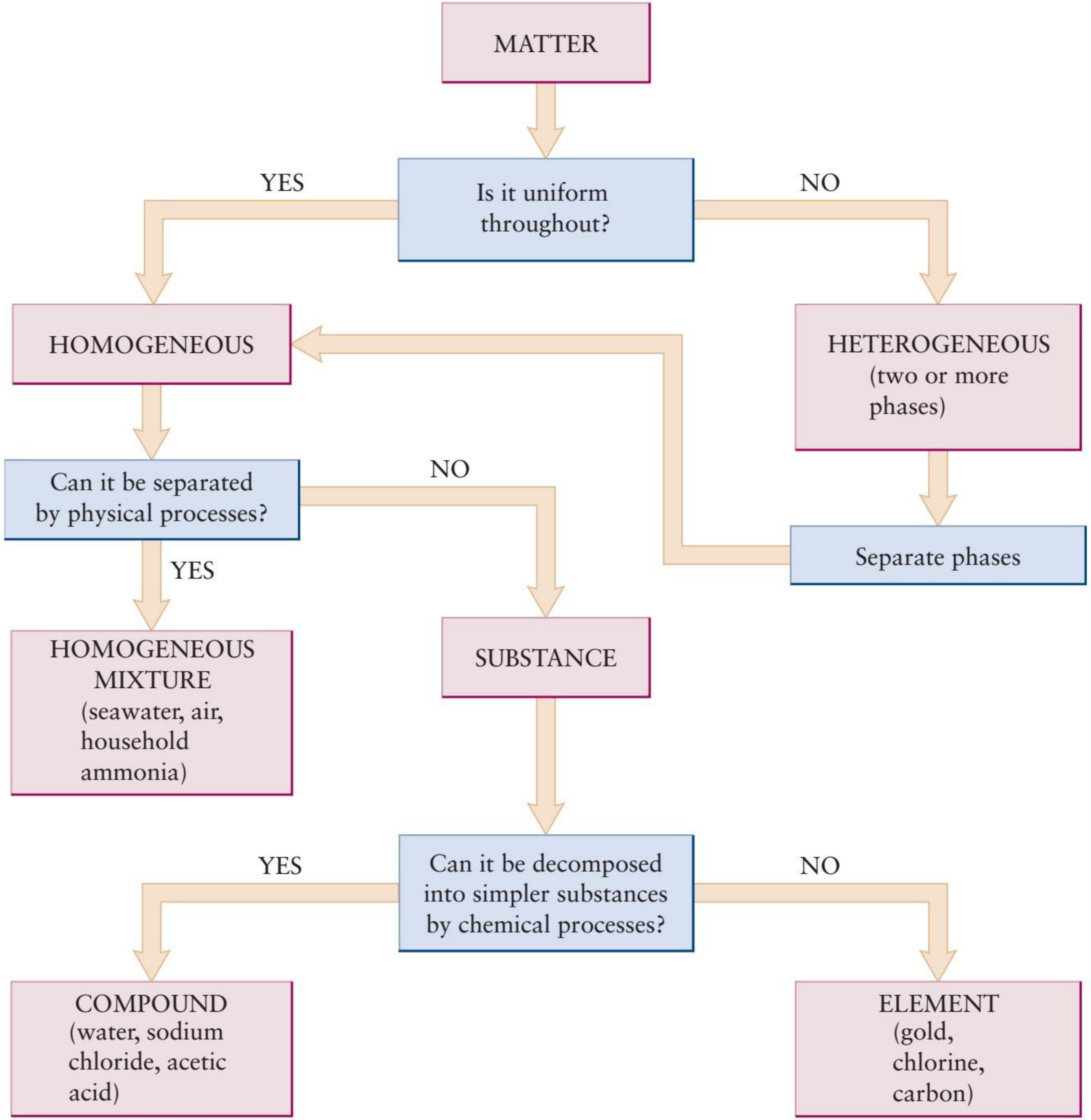


- **Mixture (混合物)**: it is possible to separate it into components by ordinary physical means (such as melting, freezing, boiling, or dissolving in solvents).

- **Substance (纯净物)**: all these physical procedures fail to separate matter into portions that have different properties.

Q1：氧气? **Q2：盐开水?**

In practice, nothing is absolutely pure, so the word substance is an idealization.

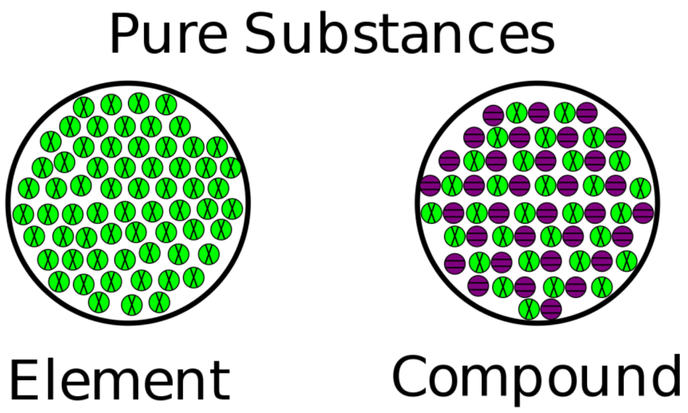


Element (单质): substance that cannot be decomposed into two or more simpler substance by ordinary physical or chemical means.

Compound (化合物) contains two or more chemical elements

Q1：氧气?

Q2：水?



Elements are building blocks of matter.

Refining Platinum

	1	2
5	4	3

Pt

Chemical Elements

Symbol

Atomic number

Relative atomic mass

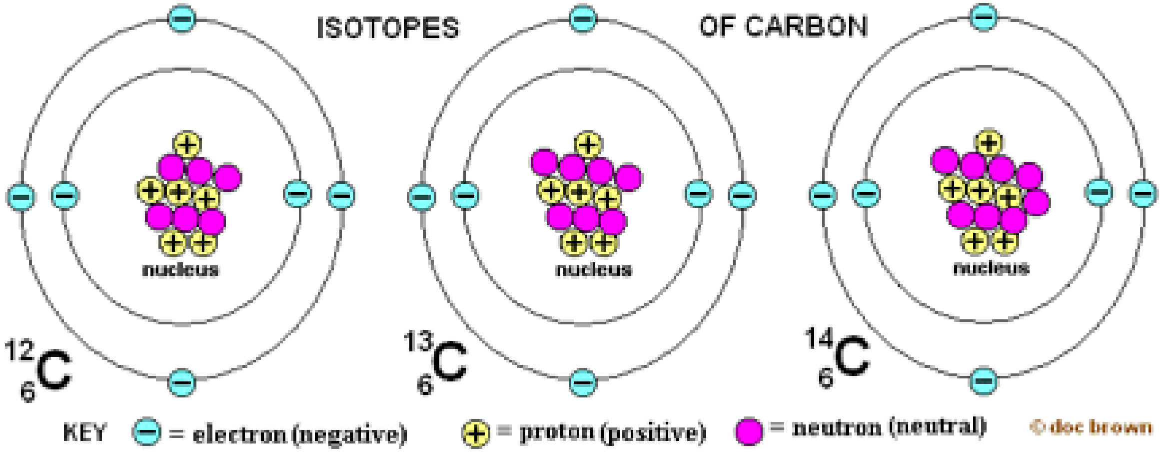
Platine
Platin
铂
白金

Name(s)

allotrope (同素异形体) and isotope (同位素)

- Graphite vs. diamond

- ^{12}C vs. ^{13}C vs. ^{14}C

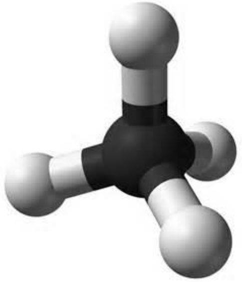


Major points of today's lecture

- **The atom in modern chemistry**
  1. **The nature of modern chemistry**
  2. **Elements: the building blocks of matter**
  3. **Laws of chemical combination**
  4. **The physical structure of atoms**
  5. **Mass spectrometer and isotopes**
  6. **The mole**

Indirect evidence of the existence of atoms

Methane

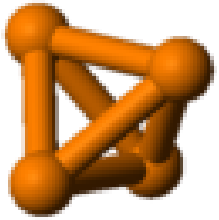


CH_4

Ethanol

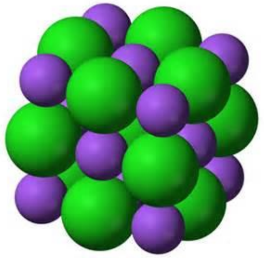
C_2H_6O

White phosphorus



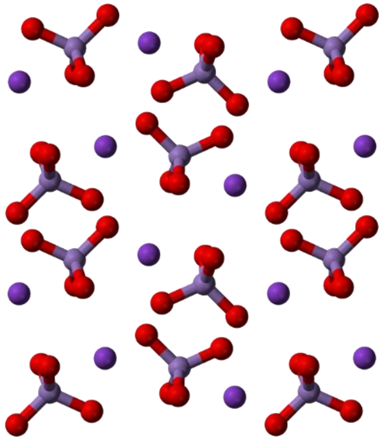
P_4

Sodium chloride



NaCl

Potassium permanganate



$KMnO_4$

How did we acquire the chemical evidence for the existence of atoms?

Laws of chemical combinations

- **The ratios of the masses of compounds react to form other compounds is fixed. As the smallest indivisible units of the compounds (atoms) combined to form the smallest indivisibles units of the compounds (moleculars).**

1. Law of Conservation of Mass

2. Law of Definite Proportions

3. Law of Multiple Proportions

4. Law of Combining Volumes

5. Avogadro's Law

Conservation of Matter and Energy

质量守恒定律：The total amount of matter involved in any chemical reaction is conserved—that is, it remains constant throughout the reaction. Matter is neither created nor destroyed in chemical reactions.

能量守恒定律：The total amount of energy involved in any chemical reaction is conserved—that is, it remains constant throughout the reaction. Energy is neither created nor destroyed in chemical reactions.

在任何与周围隔绝的物质系统（孤立系统）中，不论发生何种变化或过程，其总质量和总能量保持不变。

Q1：给手机的电池充电后，质量是否会变化？

Q2：如何减肥？

Q3：$E=mc^2$？

1. Law of Conservation of Mass

Phlogiston theory (燃素说)（early 18th century). Phlogiston was driven out of wood or metal by the heat of a fire.

Wood vs. metal?

Discovery of new elements

1. Law of Conservation of Mass

Phlogiston theory (燃素说) （early 18th century). Phlogiston was driven out of wood or metal by the heat of a fire.

Wood vs. metal?

Antoine Lavoisier
(1743–1794)

$$\text{mercury + air} \xrightarrow{350\ ^\circ\text{C}} \text{mercury oxide + remaining gas}$$

$$\text{mercury oxide} \xrightarrow{>500\ ^\circ\text{C}} \text{mercury + gas}$$

$$m(\text{mercury oxide}) = m(\text{mercury}) + m(\text{gas})$$

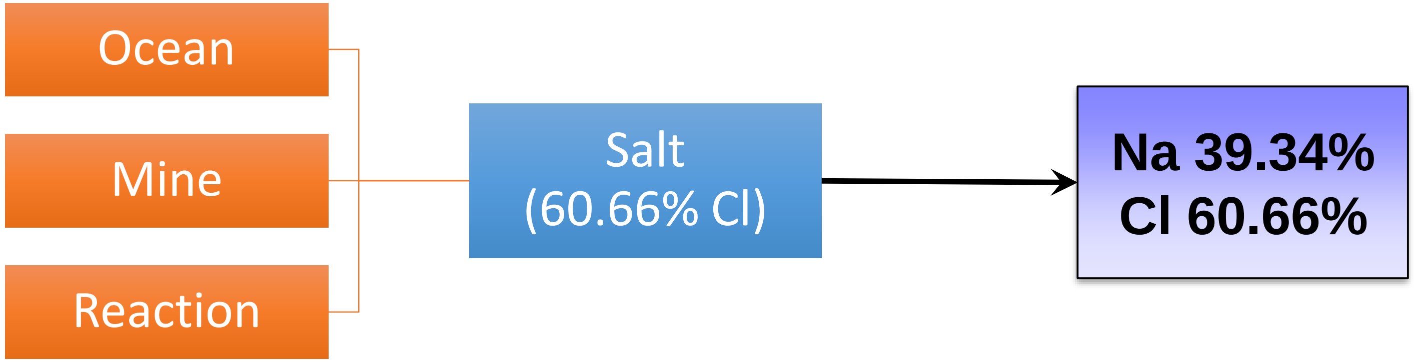
Mass was conserved during these chemical reactions

Law of conservation of mass（质量守恒定律）mass is neither created nor destroyed in chemical reactions; it is conserved.

More Elements

Symbol	Names (EN, ZH, JA)	Meaning
H	Hydrogen, 氢, 水素	Water former
O	Oxygen, 氧, 酸素	Acid former
Cl	Chlorine, 氯, 塩素	Pale green
Hg	Mercury, Hydrargyrum, 汞, 水銀	Aqueous silver
Al	Aluminum, Aluminium, 铝, アルミニウム	Alum
W	Tungsten, Wolfram, 钨, タングステン	Heavy stone
Zn	Zinc, 锌, 亜鉛	–
Sn	Tin, 锡, スズ	–

2. Law of Definite Proportions



Joseph Proust
(1754–1826)

Law of definite proportions (定比定律): in a given chemical compound, <u>the proportions by mass of the elements that compose it are fixed</u>, independent of the origin of the compound or its mode of preparation.

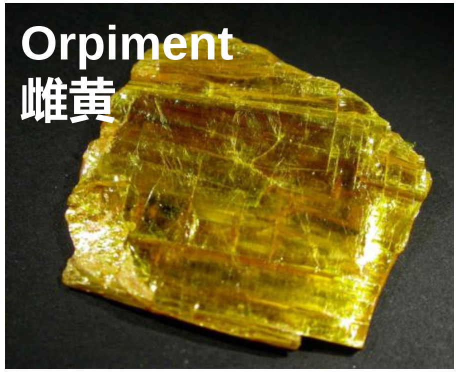
Dalton’s Atomic Theory

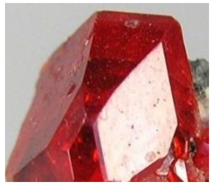
Atomic theory of matter:

1. **Matter consists of <u>indivisible atoms</u>**
2. **All <u>the atoms of a given chemical element are identical</u> in mass and in all other properties**
3. **<u>Different chemical elements</u> have different kinds of atoms; in particular, their <u>atoms have different masses</u>**
4. **Atoms are <u>indestructible</u> and retain their identities in chemical reactions**
5. **Atoms of the elements <u>combine with each other in small integer ratio</u> to form compounds**

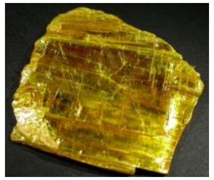
John Dalton (1766–1844)
A new system of chemical philosophy, 1808

3. Law of Multiple Proportions

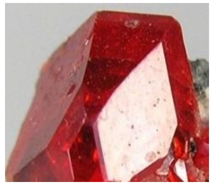




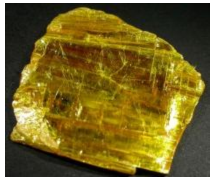
As: 70.03%
S: 29.97%



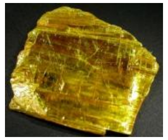
As: 60.91%
S: 39.09%



As / S
= 2.337



As / S
= 1.558



$$\left(\text{As / S} = 2.337 \right) \div \left(\text{As / S} = 1.558 \right) = 1.500 \approx \frac{3}{2}$$

Law of multiple proportions (倍比定律): when two elements form a series of compounds, the masses of one element that combine with a fixed mass of the other element are in the ratio of small integers to each other.

Empirical Formulas

For realgar "AsS", why not As_2S_2, As_3S_3, or As_4S_4?

For orpiment "As_2S_3", why not As_4S_6?

What is the formula of an element?

Hydrogen → H or H_2?

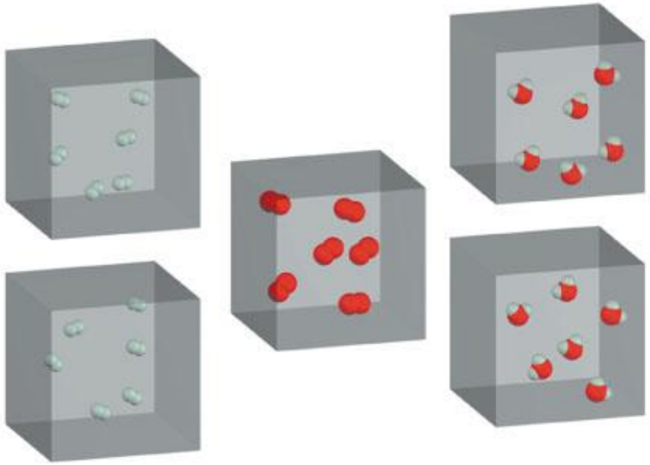
Oxygen → O, O_2, or O_3?

I prefer simplicity!

AsS and As_2S_3 are **empirical formulas**

Rule of greatest simplicity: if two elements A and B form only one compound then the chemical formula for that compound will be the simplest possible: AB. (which is incorrect)

4. Law of Combining Volumes



Joseph Louis Gay-Lussac
(1778–1850)

Hydrogen + Oxygen $\longrightarrow$ Water vapor

	Hydrogen	Oxygen	Water vapor
A_r	1	16	
Mass ratio	1	8	9
Volume ratio	2	1	2

Two independent sets of ratios

Law of combining volumes (气体化学体积定律): the ratio of the volumes of any pair of gases in a gas phase chemical reaction (at the same temperature and pressure) is the ratio of simple integers.

5. Avogadro's Hypothesis

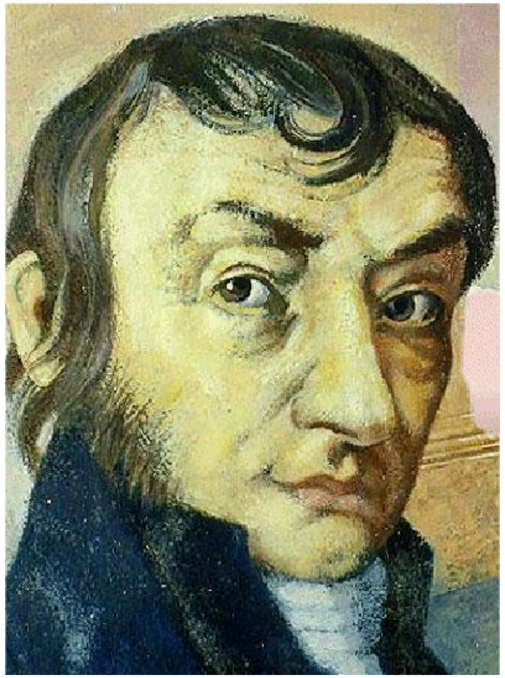
Avogadro's hypothesis (阿伏加德罗定律): equal volumes of different gases at the same temperature and pressure contain equal number of **particles**.

Dalton 1808: For all matter,

Number of **atoms** → Mass

Avogadro 1811: For all gases

Number of **particles** → Volume



Amedeo Avogadro
(1776–1856)

Particles: atoms or molecules?

5. Avogadro's Hypothesis

2 volumes of hydrogen + 1 volume of oxygen $\longrightarrow$ 2 volumes of water vapor

1 volume of nitrogen + 1 volume of oxygen $\longrightarrow$ 2 volumes of nitrogen oxide

3 volumes of hydrogen + 1 volume of nitrogen $\longrightarrow$ 2 volumes of ammonia

Avogadro

$$2\ H_2 + O_2 \longrightarrow 2\ H_2O$$

$$N_2 + O_2 \longrightarrow 2\ NO$$

$$3\ H_2 + N_2 \longrightarrow 2\ NH_3$$

Correct

Validated 50 years later

Dalton

$$H + O \longrightarrow OH$$

$$N + O \longrightarrow NO$$

$$H + N \longrightarrow NH$$

Wrong

5. Avogadro’s Hypothesis

	Hydrogen +	Oxygen $\longrightarrow$	Water vapor
Mass ratio	1	8	9
Volume ratio	2	1	2
M_r	1	16	9?

Chlorine (Cl) and oxygen form four different binary compounds. Analysis gives the following results:

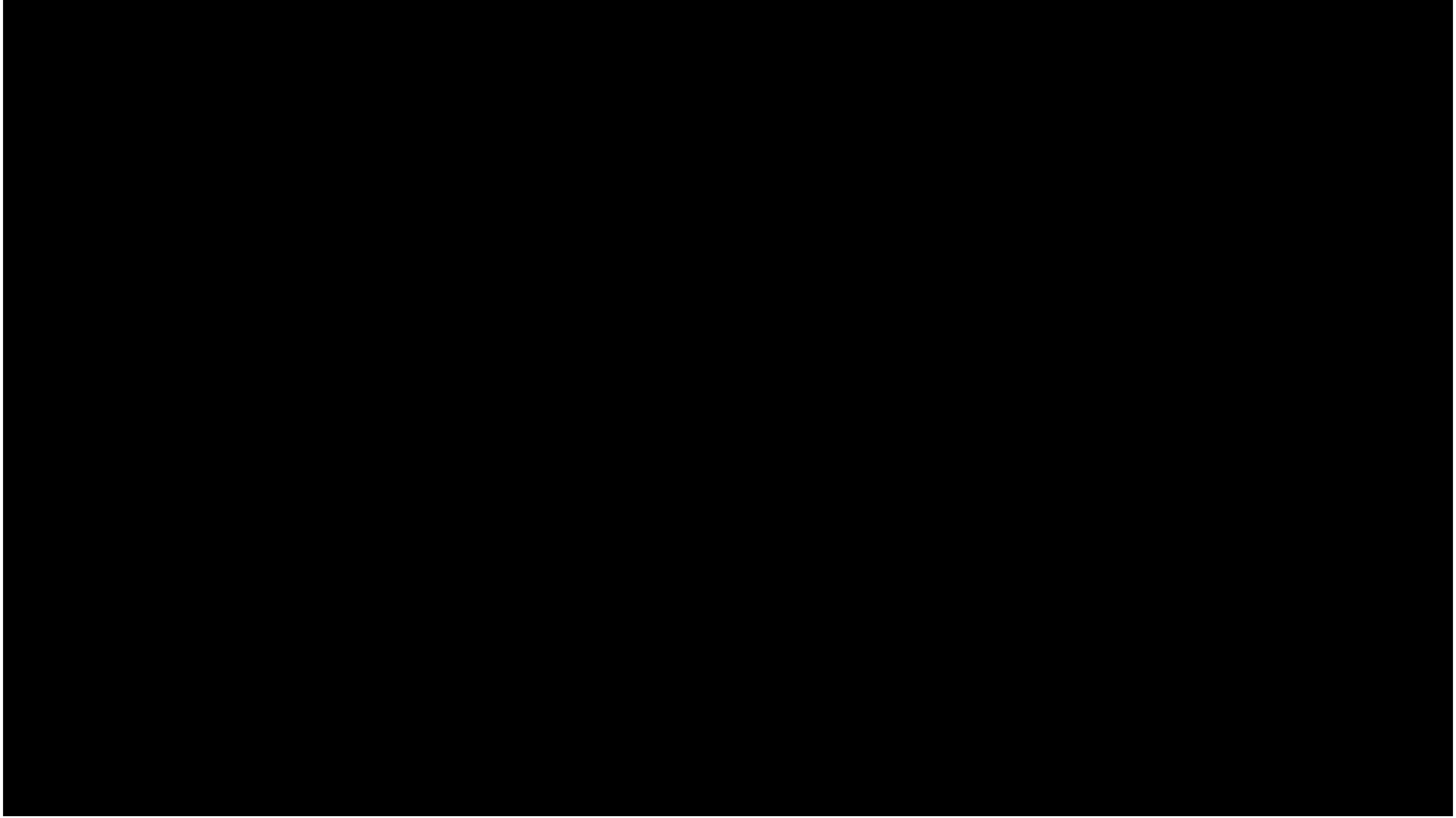
Compound	Mass of O Combined with 1.0000 g Cl
A	0.22564 g
B	0.90255 g
C	1.3539 g
D	1.5795 g

(a) Show that the law of multiple proportions holds for these compounds.

(b) If the formula of compound A is a multiple of Cl_2O, then determine the formulas of compounds B, C, and D.

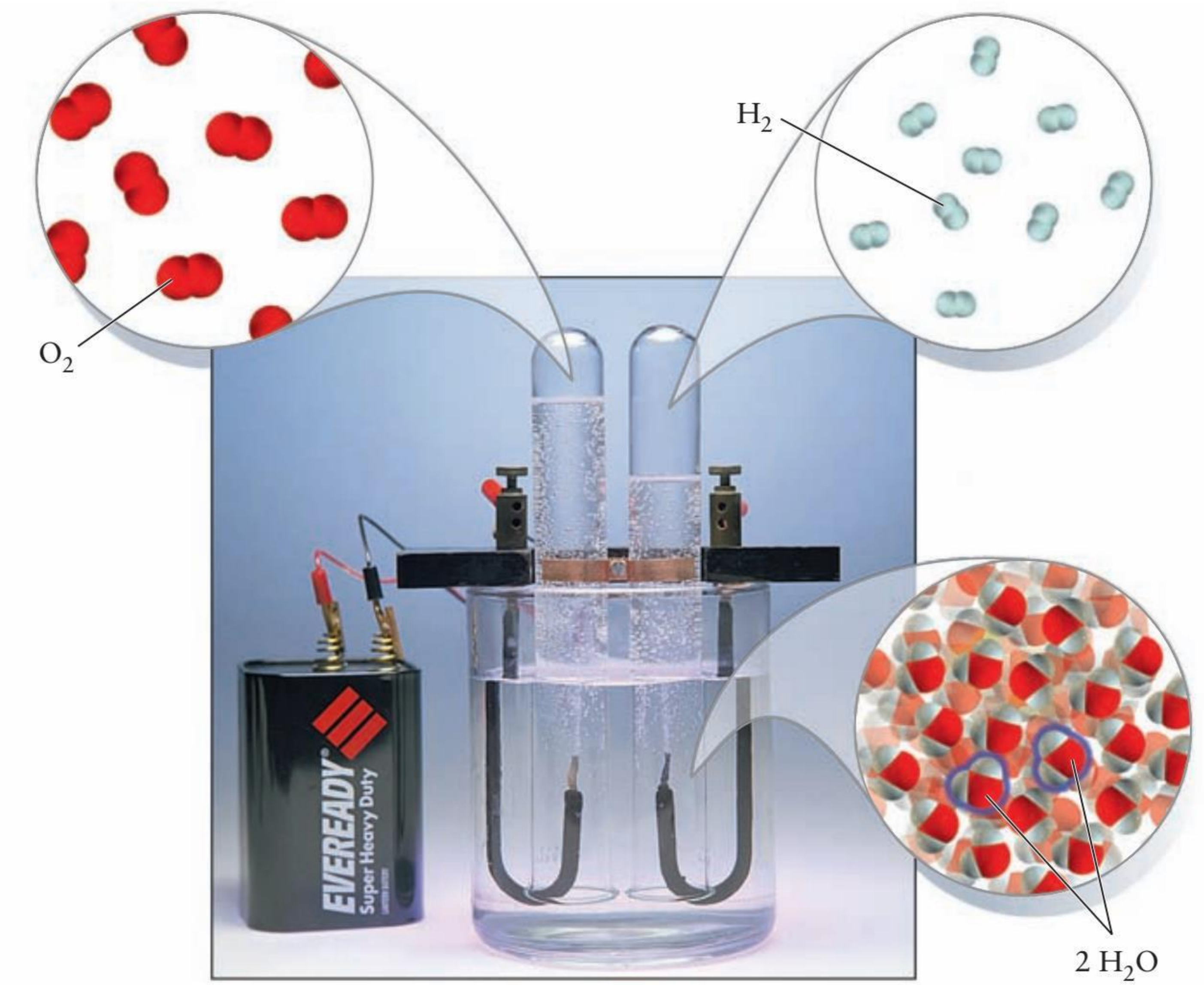
Major points of today's lecture

- **The atom in modern chemistry**
  1. **The nature of modern chemistry**
  2. **Elements: the building blocks of matter**
  3. **Laws of chemical combination**
  4. **The physical structure of atoms**
  5. **Mass spectrometer and isotopes**
  6. **The mole**



https://www.youtube.com/watch?v=oSCX78-8-q0&t

The physical structure of atoms

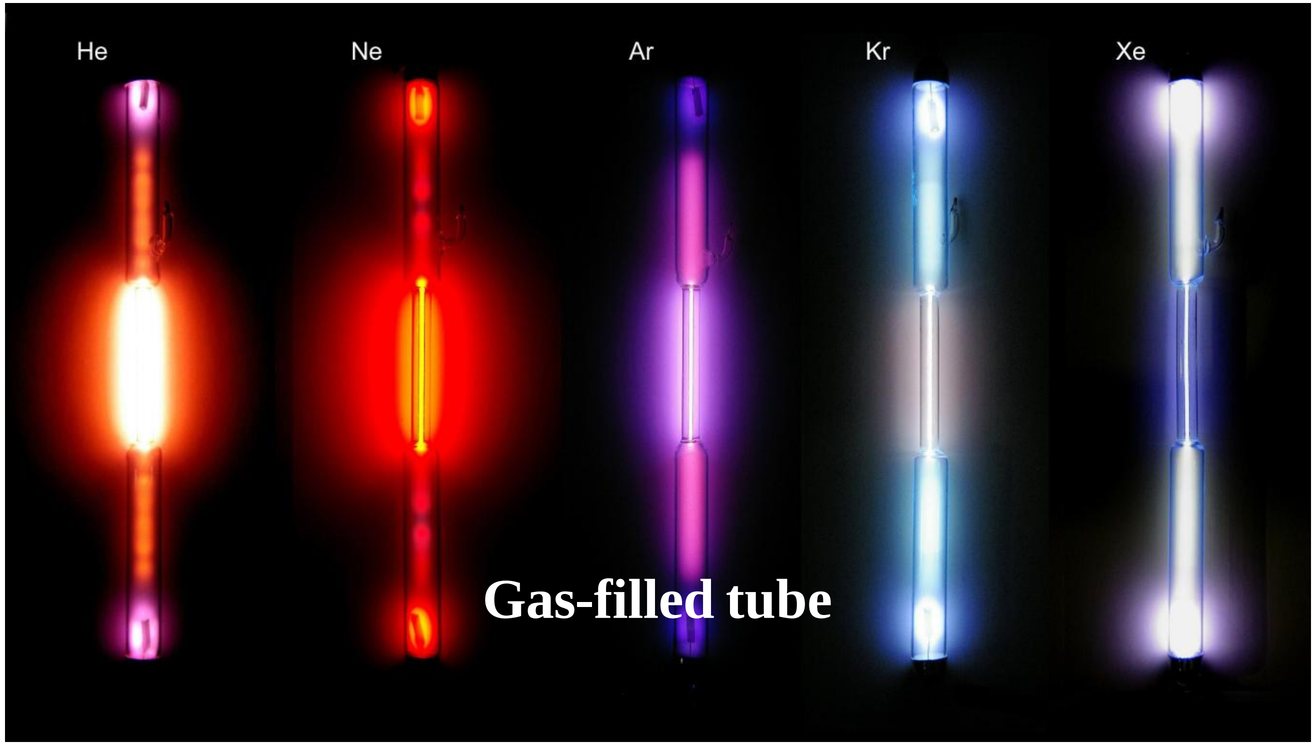


The atom was considered the ultimate and indivisible building block of matter.

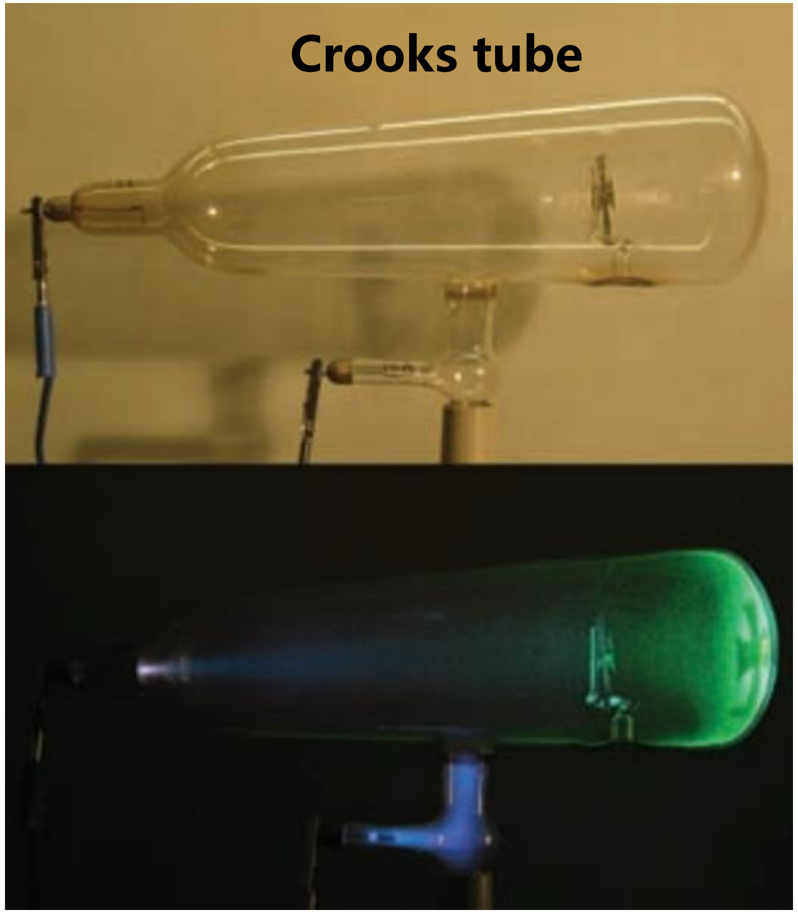
But by the end of the 19th century, this notion began to be replaced by the view that atoms were themselves composed of smaller, elementary particles.

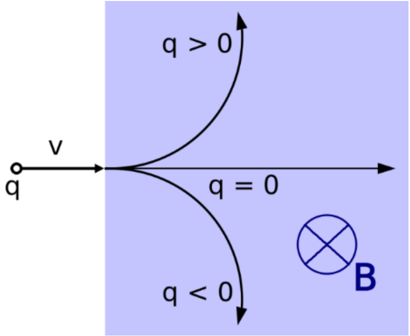
Glow discharge 辉光放电 and cathode rays 阴极射线

- **Gas discharge tubes contained gases at moderate-to-low pressure.**
- **The electric field applied between the cathode and anode of a discharge tube caused "electrical breakdown" of the gas to form a *glow discharge* that emitted light.**



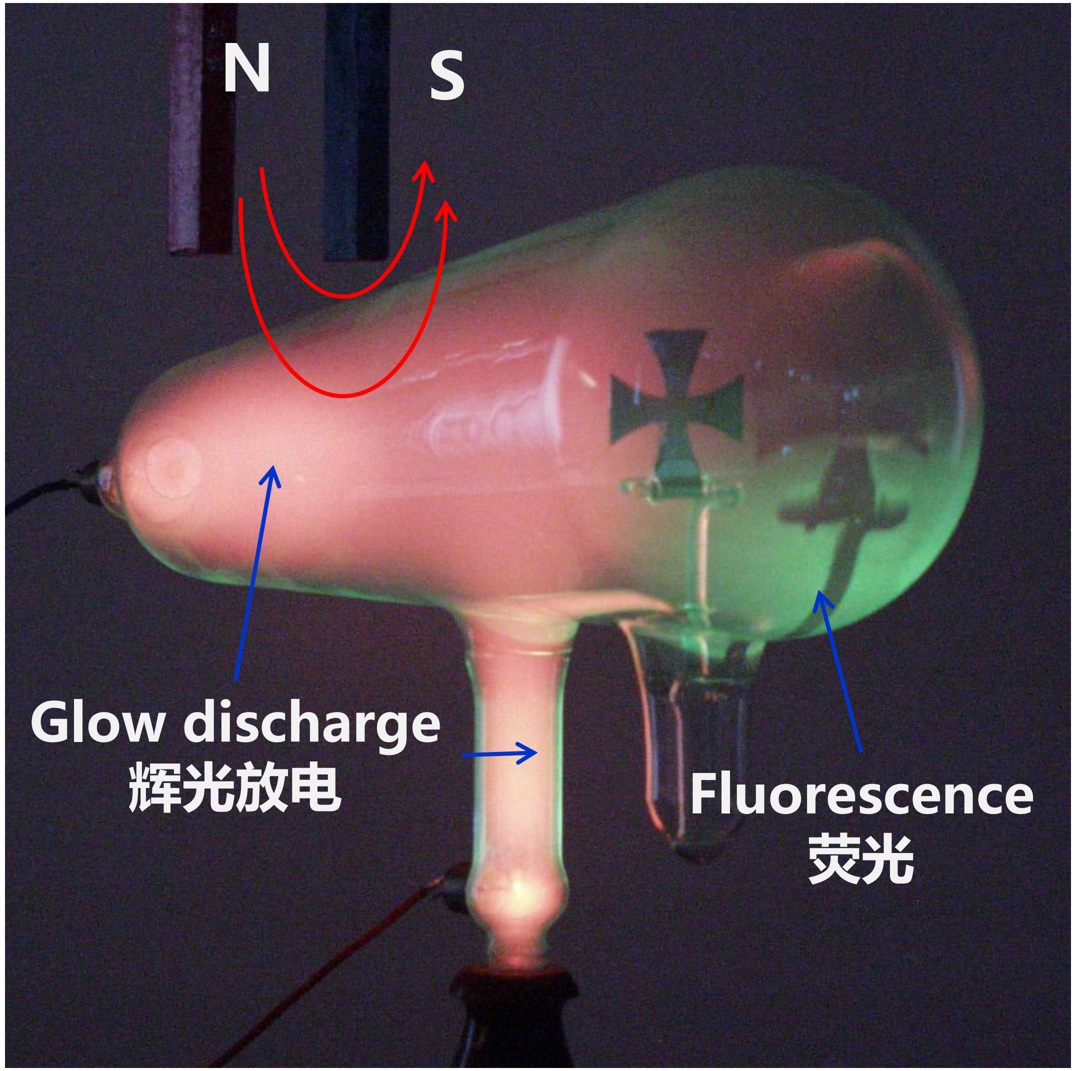
Glow discharge and cathode rays





Lorentz force

Particles in the cathode ray are negatively charged.



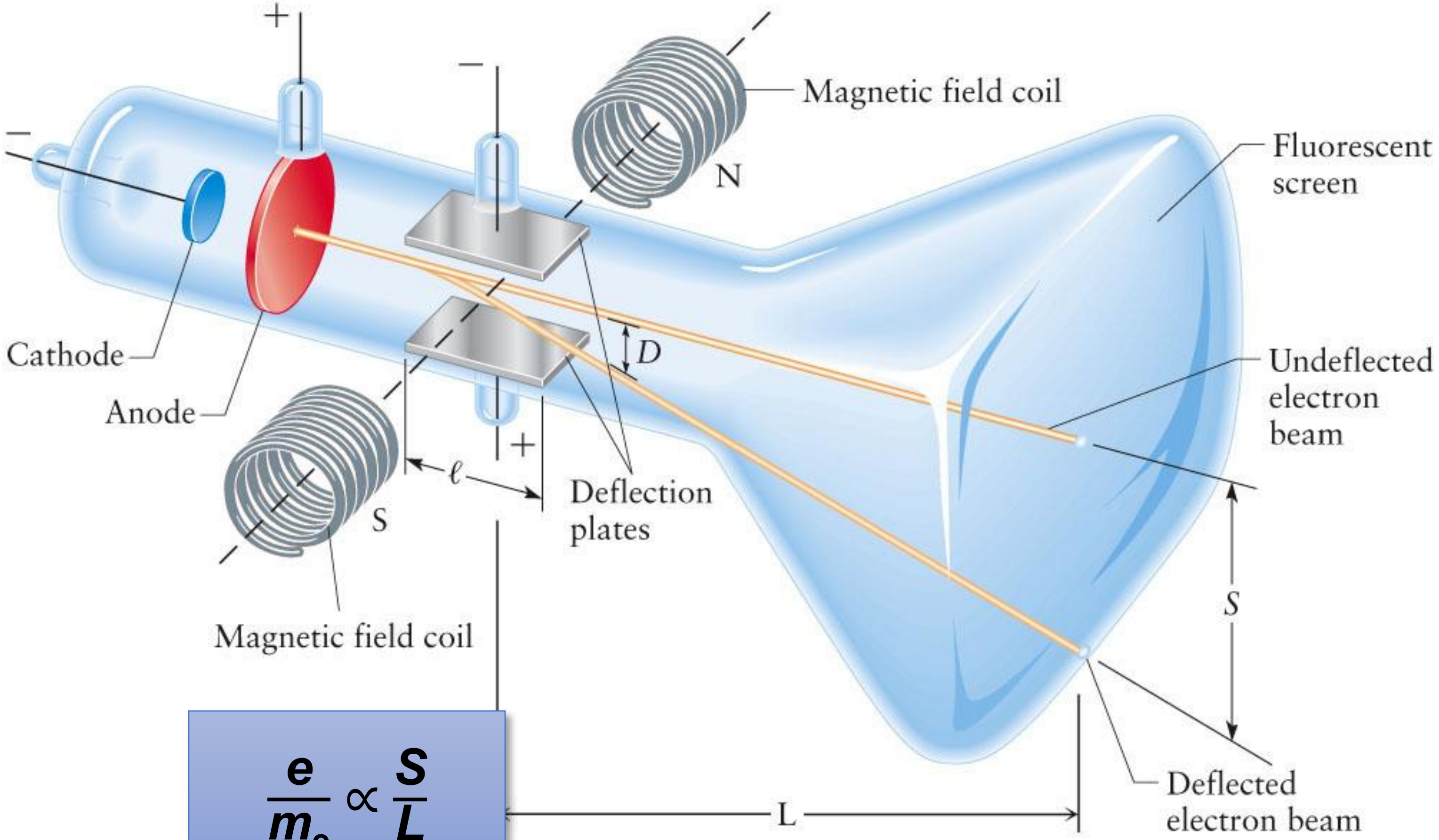
Tubes de Crookes

(1)

Tube à croix de Malte

https://www.youtube.com/watch?v=IVTA6moqVEM

Cathode Ray Tube (CRT)



$$\frac{e}{m_e} \propto \frac{S}{L}$$

J. J. Thomson (1856–1940)

$\frac{e}{m_e}$ **is independent of the cathode material or the gas.**

The Plum Pudding Model

J. J. Thomson

- **1897 Discovered electron**
- **1904 Proposed the pudding model**

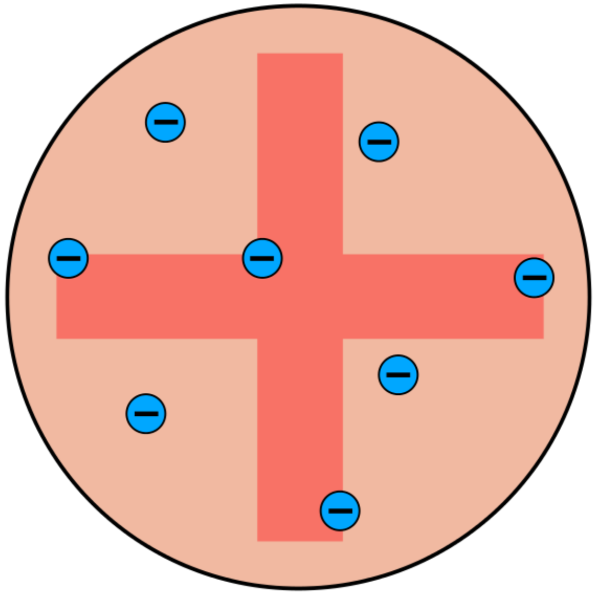


Photo from the Nobel Foundation archive.

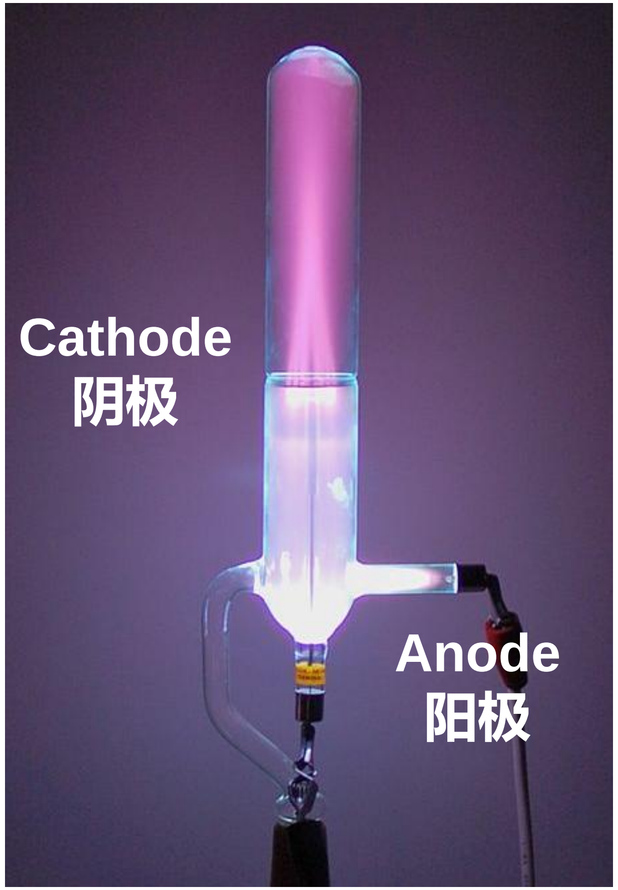
Joseph John Thomson

Prize share: 1/1

The Nobel Prize in Physics 1906 **was awarded to Joseph John Thomson "<u>in recognition of the great merits of his theoretical and experimental investigations on the conduction of electricity by gases</u>".**

https://www.nobelprize.org/prizes/physics/1906/summary/

Canal Rays 阳极射线

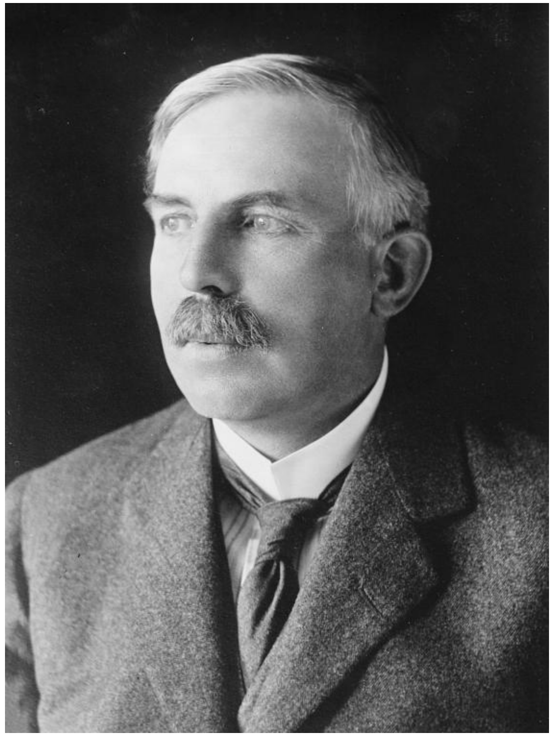


The particles that emerged from the glow discharge to form canal rays were accelerated toward the cathode: positively charged.

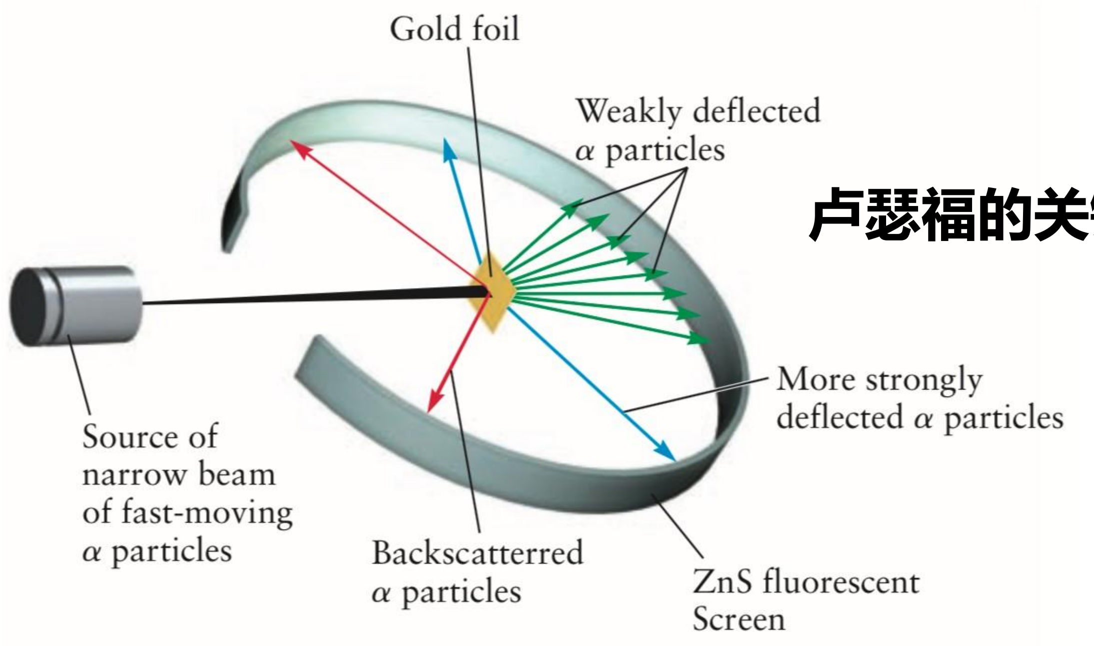
Much larger electric and magnetic fields were required to deflect these particles than those used in Thomson's experiments: more massive than the electron.

If different gases were leaked into the apparatus, the magnitude of the fields required to displace canal rays of a different gas by the same amount differed: charged particles associated with each gas had different masses.

Rutherford's Gold Foil Experiment (1911)



Ernest Rutherford (1871-1937)

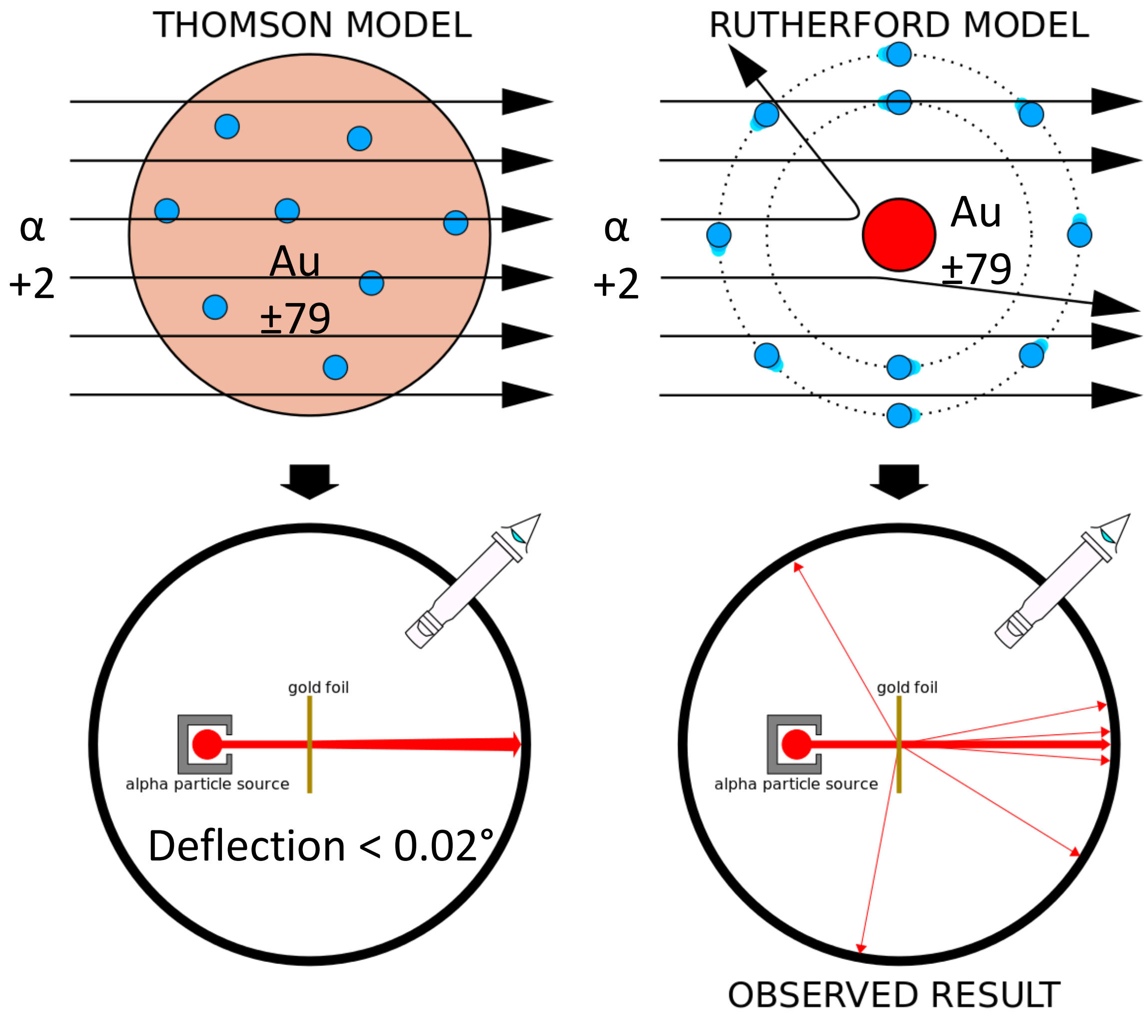


卢瑟福的关键发现

Almost all of the alpha particles passed straight through the foil, but a few were deflected through large angles.

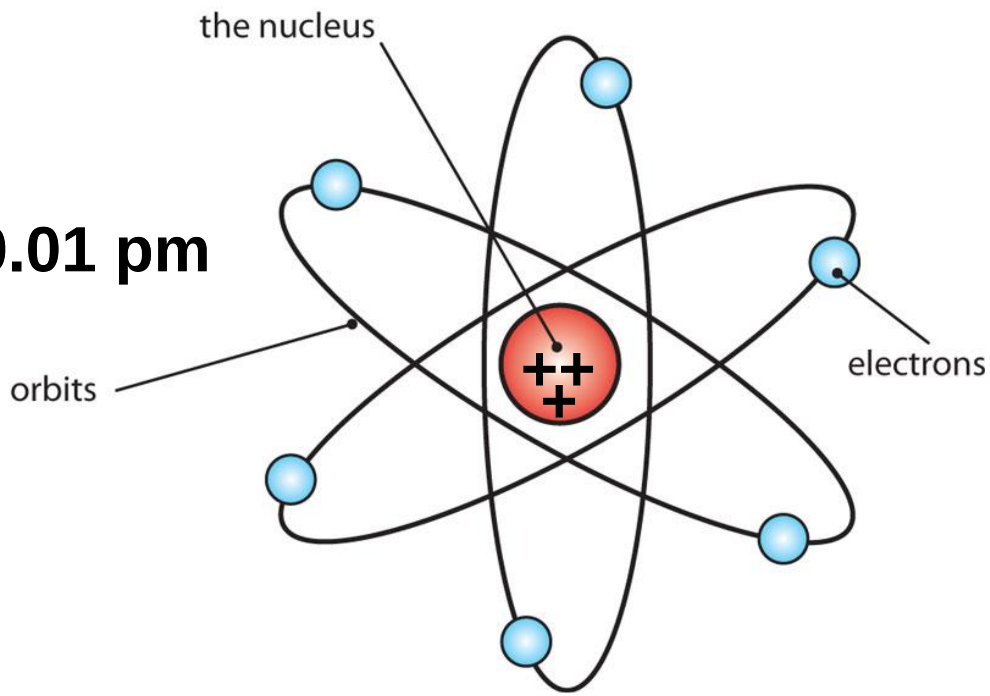
Rutherford's Gold Foil Experiment (1911)

Rutherford:" it was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you."



The Rutherford Model

< 0.01 pm



← ~100 pm →

IAEA

International Atomic Energy Agency

The Au nucleus has 79 positive charges
but weighs 197 mass units?

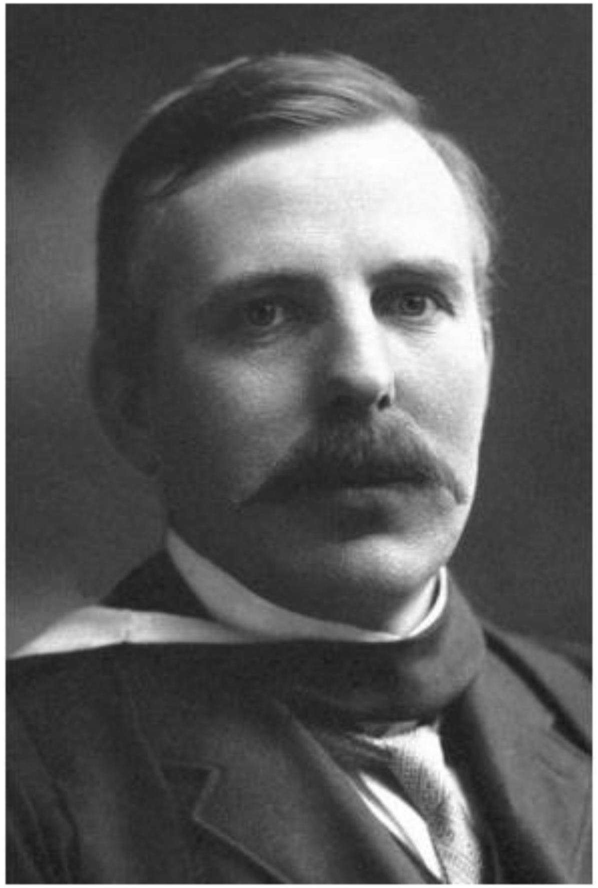


Photo from the Nobel Foundation archive.

Ernest Rutherford

Prize share: 1/1

The Nobel Prize in Chemistry 1908 was awarded to Ernest Rutherford "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances"

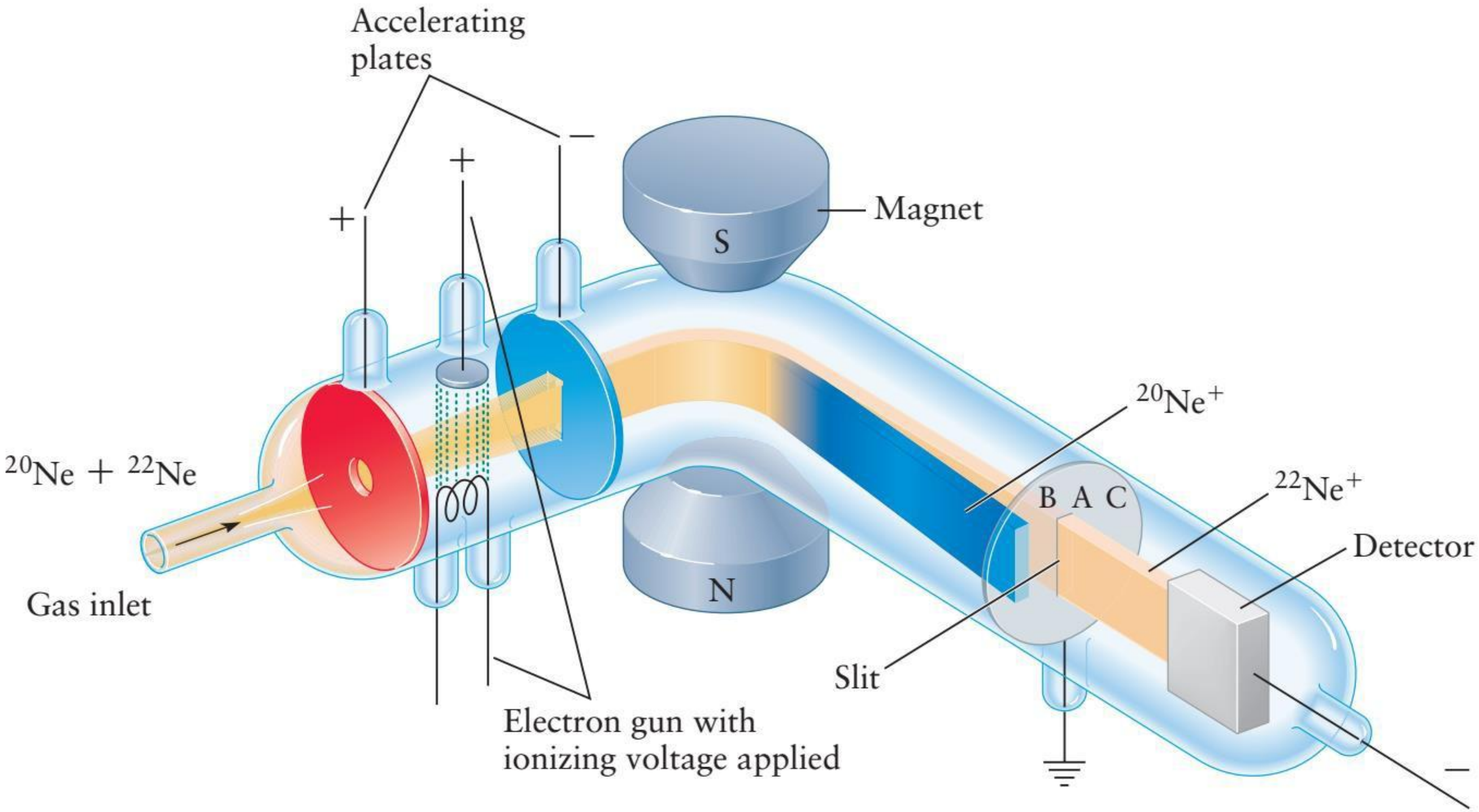
https://www.nobelprize.org/prizes/chemistry/1908/summary/

Major points of today's lecture

- The atom in modern chemistry

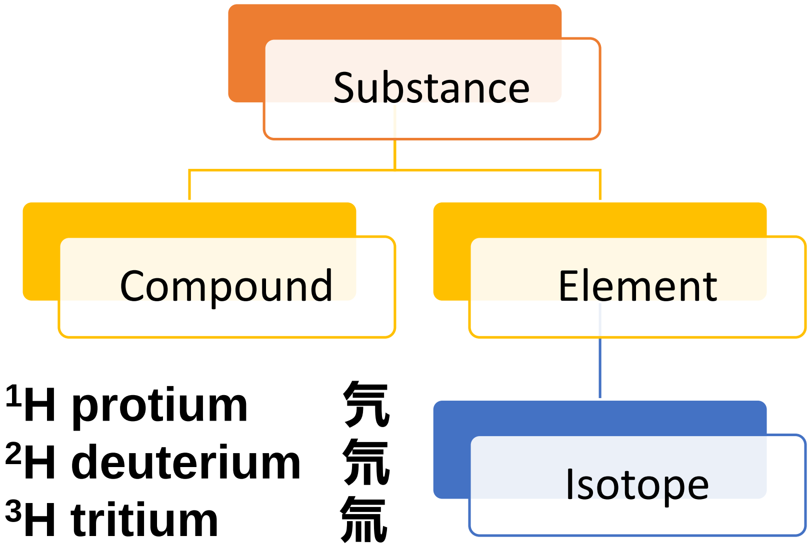
1. The nature of modern chemistry
2. Elements: the building blocks of matter
3. Laws of chemical combination
4. The physical structure of atoms
5. Mass spectrometer and isotopes
6. The mole

Mass spectroscopy

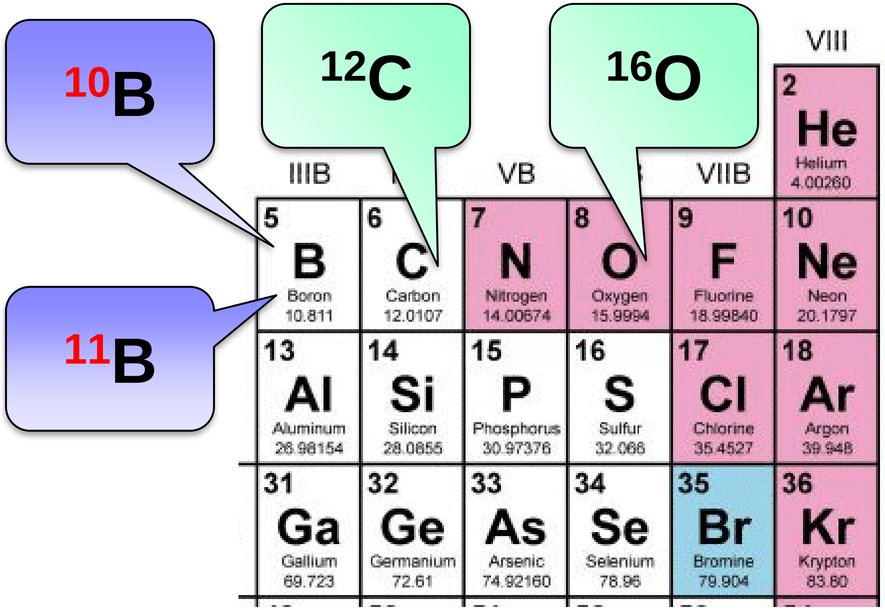


Relative Atomic Mass and Isotope

	H	B	C	O	Mg
A_r	1.0	10.8	12.0	16.0	24.3



^{1}H protium 氕
^{2}H deuterium 氘
^{3}H tritium 氚



Isotope 1	Abundance	Isotope 2	Abundance	A_r
^{10}B or B-10	~20%	^{11}B or B-11	~80%	B = 10.8
^{35}Cl or Cl-35	~75%	^{37}Cl or Cl-37	~25%	Cl = 35.5

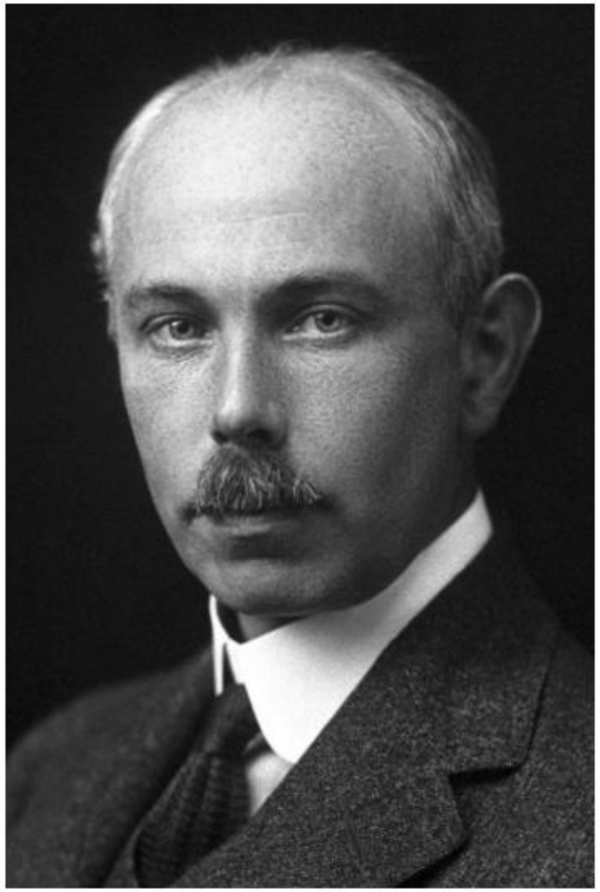


Photo from the Nobel Foundation archive.

Francis William Aston

Prize share: 1/1

The Nobel Prize in Chemistry 1922 was awarded to Francis William Aston "for his discovery, by means of his mass spectrograph, of isotopes, in a large number of non-radioactive elements, and for his enunciation of the whole-number rule"

https://www.nobelprize.org/prizes/chemistry/1922/summary/

Relative Atomic Mass and Isotope

H	B	C	O	F
1.008 99.98%	10.013 80%	12 98.9%	15.995 99.76%	18.998 100%
2.014 0.02%	11.009 20%	13.003 1.1%	16.999 0.04%	
			17.999 0.2%	

1808 **Dalton proposed $A_r(H) = 1$**

~1900 **$A_r(O) = 16$**

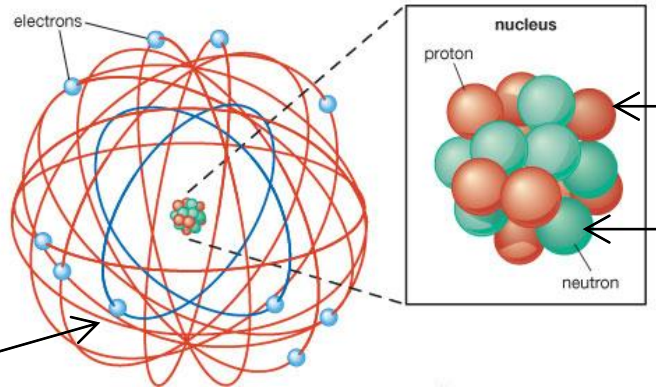
1961 **Physicists & chemists agreed $A_r(^{12}C) = 12$**

Discovery of Neutron

1920 Rutherford proposed

a proton-electron model

for the nucleus



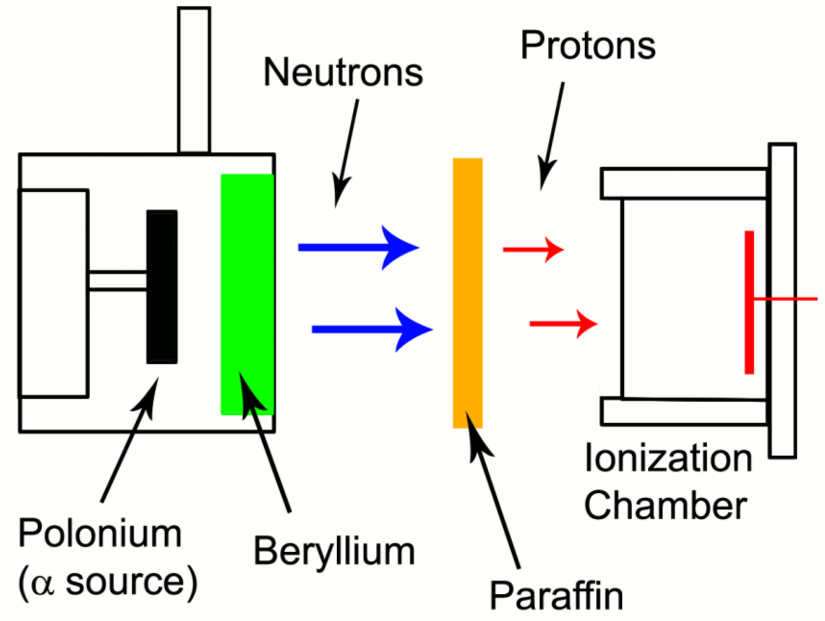
79 protons

118 neutrons

79 electrons

1932 Chadwick

discovered the neutron



James Chadwick
(1891–1974)

Photo from the Nobel Foundation archive.

James Chadwick

Prize share: 1/1

The Nobel Prize in Physics 1935 was awarded to James Chadwick "for the discovery of the neutron".

https://www.nobelprize.org/prizes/physics/1935/summary/

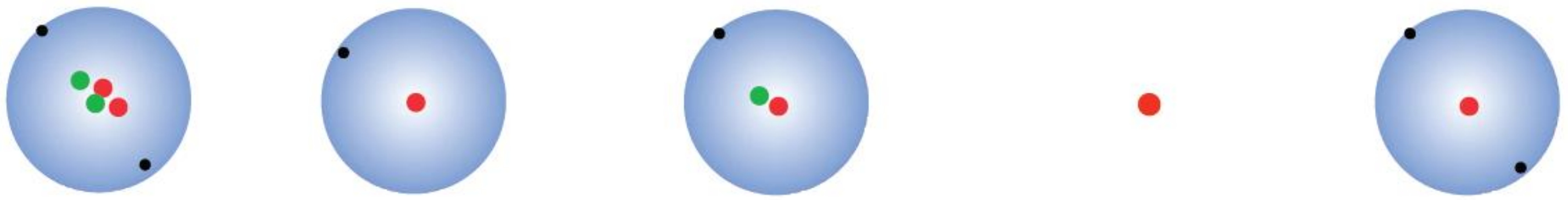
Summary

Mass number
A = p + n

Atomic number (charge number)
Z = p

$$^{A}_{Z}\mathrm{X}$$

Anode	阳极
Anion	阴离子/负离子
Cathode	阴极
Cation	阳离子/正离子



Helium, $^{4}_{2}\mathrm{He}$ — Protium, $^{1}_{1}\mathrm{H}$ — Deuterium, $^{2}_{1}\mathrm{H}$ — Hydrogen ion (proton), H^{+} — Hydride, H^{-}

Cation — Anion

$\underbrace{\text{Protium, } ^{1}_{1}\mathrm{H} \quad \text{Deuterium, } ^{2}_{1}\mathrm{H}}_{\text{Hydrogen, } _{1}\mathrm{H}}$ $\quad \underbrace{\text{Cation} \quad \text{Anion}}_{\text{Ion}}$

$\underbrace{\text{Hydrogen, } _{1}\mathrm{H} \quad \text{Ion}}_{\text{Generalized hydrogen element 元素}}$

Generalized hydrogen element 元素

Calculate the relative atomic mass of carbon, taking the relative atomic mass of ^{13}C to be 13.003354 on the ^{12}C scale.

Solution

Set up the following table:

Isotope	Isotopic Mass × Abundance
^{12}C	$12.000000 \times 0.98892 = 11.867$
^{13}C	$13.003354 \times 0.01108 = 00.144$

Chemical relative atomic mass = 12.011

Radon-222 (^{222}Rn) has recently received publicity because its presence in basements may increase the number of cancer cases in the general population, especially among smokers. State the number of electrons, protons, and neutrons that make up an atom of ^{222}Rn.

Solution

From the table on the inside front cover of the book, the atomic number of radon is 86; thus, the nucleus contains 86 protons and $222 - 86 = 136$ neutrons. The atom has 86 electrons to balance the positive charge on the nucleus.

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The Mole: Weighing and Counting Molecules

- 19th century chemists concluded that oxygen atoms weigh 16 times as much as hydrogen atoms.
- In the early 20th century, relative atomic and molecular masses were determined much more accurately through the work of J. J. Thomson, F. W. Aston, and others using mass spectrometry.
- These relative masses on the atomic scale must be related to absolute masses on the gram scale by a conversion factor.

How to weigh out a sample containing exactly the number of atoms or molecules needed for a particular chemical reaction?

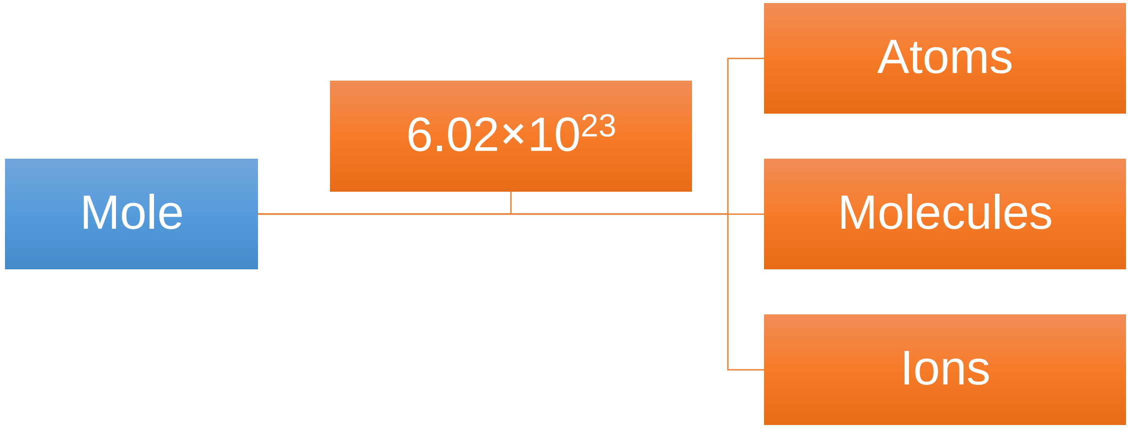
微观原子质量与 vs. 观物质质量的关系

The Mole: Weighing and Counting Molecules

- **Avogadro's number (阿伏伽德罗数N_A):** The number of atoms in exactly 12g of ^{12}C. $N_A = 6.02214120 \times 10^{23}$ （principle of modern chemistry, 7th edition）

$$N \equiv \frac{12\ \text{g}}{m_{a}(^{12}\text{C})} = \frac{12\ \text{g}}{1.99\times10^{-23}\ \text{g}} = 6.02\times10^{23}$$

- **The mole (摩尔数):** One mole of a substance is the amount that contains Avogadro's number of atoms, molecules, or other entities



1 mol of ^{12}C contains N_A ^{12}C atoms, 1 mol of water contains N_A water molecules.

- **The molar Mass (摩尔质量):** one mole of that a substance in grams. Molar mass: M (g/mol)

$A_r(\text{Ag}) = 107.9 \quad \rightarrow M(\text{Ag}) = 107.9\ \text{g mol}^{-1}$.

$M_r(H_2O) = 18.0 \quad \rightarrow M(H_2O) = 18.0\ \text{g mol}^{-1}$.

THE SI

The SI — the modern metric system — has seven base units from which all other measurement units can be derived. On May 20, 2019, four of them — the kilogram, kelvin, ampere and mole — were redefined in terms of constants of nature. The remaining three — the second, meter, and candela — are already based on universal constants.

Click on the SI symbols below for more information.

Kilogram
kg

Candela
cd

Kelvin
K

Meter
m

Second
s

Ampere
A

Mole
mol

Mole – Unit of Amount of Substance

Symbol: mol

One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant, N_A, when expressed in the unit mol^{-1} and is called the Avogadro number. The amount of substance, symbol *n*, of a system is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle or specified group of particles.

Learn More

Length: how long is one meter?

- Y1791: one ten-millionth of the length of a great circle quadrant along the Earth's meridian through Paris

- Y1799: platinum meter bar

- Y1983: the length of the path traveled by light in a vacuum in 1/299,792,458 of a second

“坐地日行八万里，巡天遥看一千河”

One of the heaviest atoms found in nature is ^{238}U. Its relative atomic mass is 238.0508 on a scale in which 12 is the atomic mass of ^{12}C. Calculate the mass (in grams) of one ^{238}U atom.

Solution

Because the mass of N_A atoms of ^{238}U is 238.0508 g and N_A is 6.0221420×10^{23}, the mass of one ^{238}U atom must be

$$\frac{238.0508 \text{ g}}{6.0221420 \times 10^{23}} = 3.952926 \times 10^{-22} \text{ g}$$

Nitrogen dioxide (NO_2) is a major component of urban air pollution. For a sample containing 4.000 g NO_2, calculate (a) the number of moles of NO_2 and (b) the number of molecules of NO_2.

Solution

(a) From the tabulated molar masses of nitrogen (14.007 g mol^{-1}) and oxygen (15.999 g mol^{-1}), the molar mass of NO_2 is

$$14.007 \text{ g mol}^{-1} + (2 \times 15.999 \text{ g mol}^{-1}) = 46.005 \text{ g mol}^{-1}$$

The number of moles of NO_2 is then

$$\text{mol of NO}_2 = \frac{4.000 \text{ g NO}_2}{46.005 \text{ g mol}^{-1}} = 0.08695 \text{ mol NO}_2$$

(b) To convert from moles to number of molecules, multiply by Avogadro's number:

$$\text{molecules of NO}_2 = \left(0.08695 \text{ mol NO}_2\right) \times 6.0221 \times 10^{23} \text{ mol}^{-1}$$
$$= 5.236 \times 10^{22} \text{ molecules NO}_2$$

Density and Molecular Size

The density of a sample is the ratio of its mass to its volume:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

单位： kg/m^3

TABLE 2.1 Densities of Some Substances

Substance	Density ($g\ cm^{-3}$)
Hydrogen	0.000082
Oxygen	0.00130
Water	1.00
Magnesium	1.74
Sodium chloride	2.16
Quartz	2.65
Aluminum	2.70
Iron	7.86
Copper	8.96
Silver	10.5
Lead	11.4
Mercury	13.5
Gold	19.3
Platinum	21.4

These densities were measured at room temperature and at average atmospheric pressure near sea level.

Density and Molecular Size

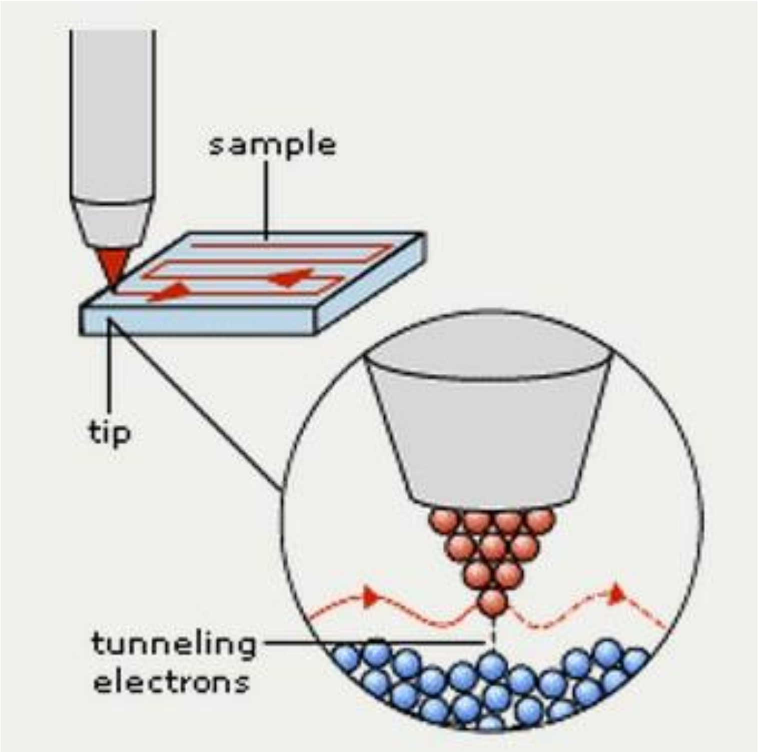
Molar volume (摩尔体积): the volume occupied by one mole of a substance:

$$V_m = \frac{\text{molar mass}\ \ (\text{g mol}^{-1})}{\text{density}\ (\text{g cm}^{-3})} = \text{molar volume}\ (\text{cm}^3\ \text{mol}^{-1})$$

For example, near 0°C, ice has a density of 0.92 g cm^{-3}; thus, the molar volume of solid water under these conditions is

$$V_m = \frac{18.0\ \text{g mol}^{-1}}{0.92\ \text{g cm}^{-1}} = 20\ \text{cm}^3\ \text{mol}^{-1}$$

Scanning Tunneling Microscopy




Gerd Binnig


Heinrich Rohrer

Phys. Rev. Lett., **1982**, *49*, 57.

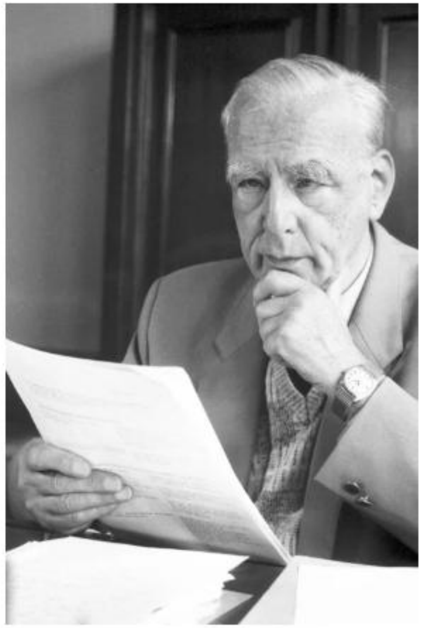


Photo from the Nobel Foundation archive.

Ernst Ruska

Prize share: 1/2

Photo from the Nobel Foundation archive.

Gerd Binnig

Prize share: 1/4

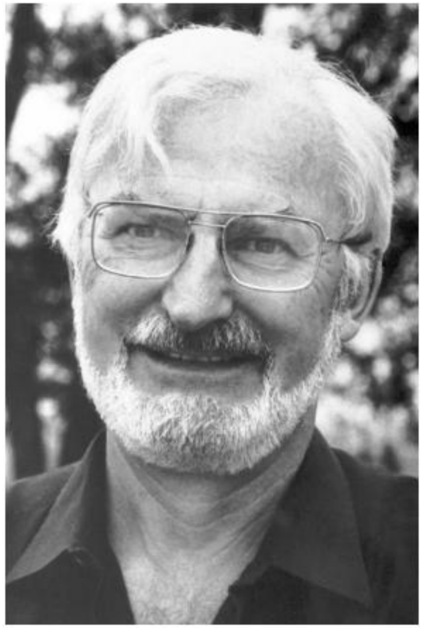


Photo from the Nobel Foundation archive.

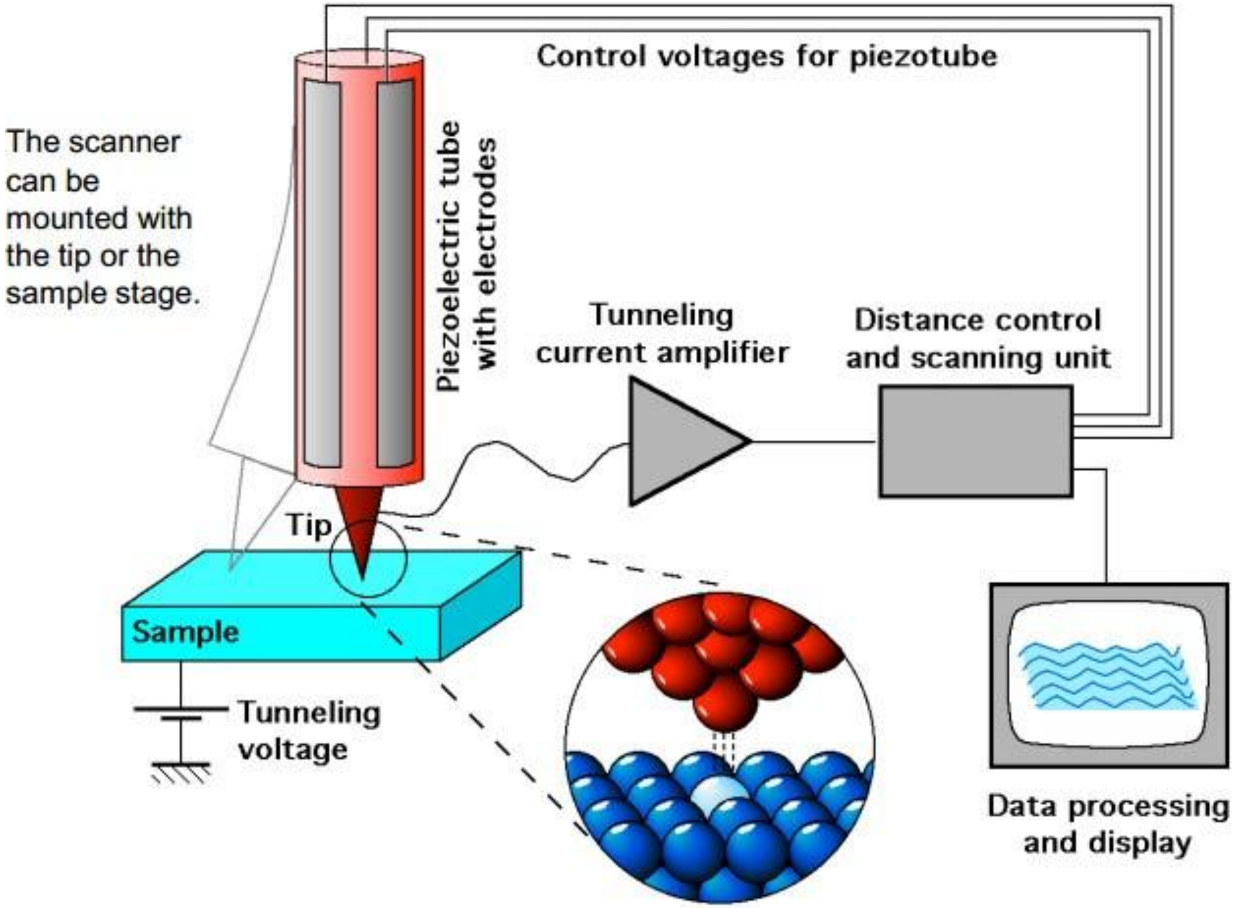
Heinrich Rohrer

Prize share: 1/4

The Nobel Prize in Physics 1986 was divided, one half awarded to Ernst Ruska "for his fundamental work in electron optics, and for the design of the first electron microscope", the other half jointly to Gerd Binnig and Heinrich Rohrer "for their design of the scanning tunneling microscope".

https://www.nobelprize.org/prizes/physics/1986/summary/

Anatomy of an STM



Five basic components:

1. Metal tip,
2. Piezoelectric scanner,
3. Current amplifier (nA),
4. Bipotentiostat (bias),
5. Feedback loop (current).

- Tunneling current from tip to sample or vice-versa depending on bias;
- Current is exponentially dependent on distance;
- Raster scanning gives 2D image;
- Feedback is normally based on constant current, thus measuring the height on surface.

https://www.youtube.com/watch?v=wNEqRq6NyUw

Homework: read “principle of modern chemistry”, chapter 1.